1 Basic Concepts

1.1 INTRODUCTION

A mechanism is a mechanical device for transferring motion and/or force from an input source to an output. Understanding how a particular mechanism works can be relatively easy, but comprehending how it originated and why it was designed in the particular form in which it exists is more difficult. The fundamental task of conceptualizing mechanisms is still a mixture of art and science, intuition and engineering. Before we introduce the basic concepts of kinematics (the motion) of mechanism design, let us consider a few mechanisms that we come across every day. Figure 1.1 shows a motorcycle suspension mechanism. The purpose of this mechanism is to guide the wheel up and down in a controlled way as it isolates the rider from the motion and vibration transmitted by the uneven road surface. This assembly is described as a four-link mechanism, consisting of three moving links and one fixed link corresponding (rigidly connected) to the frame of the motorcycle, is designed to effectively transmit the wheel motion to the shock absorber (piston and cylinder). The link dimensions are carefully determined to provide the appropriate travel of the piston inside the shock absorber in response to the motion of the wheel. The isolated suspension mechanism is diagrammed in Figure 1.1c, revealing a kinematic chain of six distinct links. Links 1 through 4 make up the four-link mechanism while links 5 and 6 represent the shock absorber.

Mechanism: a mechanical device for transferring motion and/or force from an input to an output.

Actuator: a component which uses energy to produce motion.

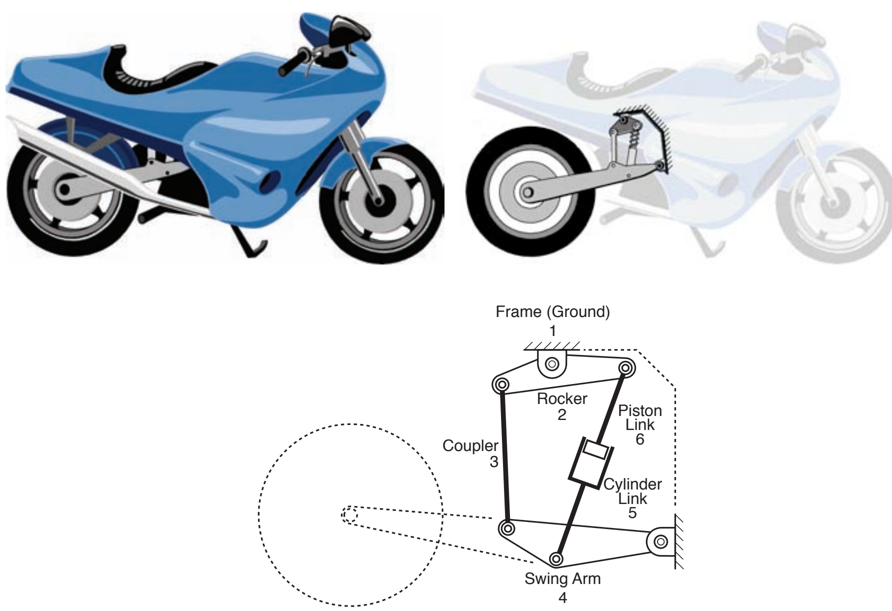


Figure 1.1 A motorcycle suspension mechanism and its kinematic structure. The four-link mechanism consists of (i) a fixed or "ground" link (frame of the motorcycle), (ii) a link (#2) to which the shock-absorber is connected, (iii) a swing arm (#4) to which the wheel is mounted, and (iv) a coupler link (#3) connecting the swing arm to link #2. The shock absorber consisting of a cylinder and piston is counted as two additional links making a total of six links in the kinematic chain.

The following are six more examples of moderately complex mechanisms that are commonly encountered. Do not be concerned if you cannot understand how the mechanisms shown in this chapter accomplish the transfer of motion between the actuator and the rest of the mechanism. This text will help you understand linkage, cam and gear systems and how to model, design and analyze them.

Figure 1.2 shows a model and schematic drawing of a prosthetic knee mechanism, another example of a mechanism designed to guide one rigid body with respect to another. In this mechanism link 1 is attached to a socket which serves as an interface between the individual's thigh and the knee mechanism. Link 3 connects the knee mechanism to the remainder of the individual's prosthetic limb, the shank and the foot. Links 2, 4, 5, and 6 produce the desired motion between links 1 and 3. The mechanism is designed to mimic the motion of a natural knee joint, which is as you can see quite unlike a simple hinge.

Differential:
a mechanism that transmits and coordinates multiple motions from a single force.

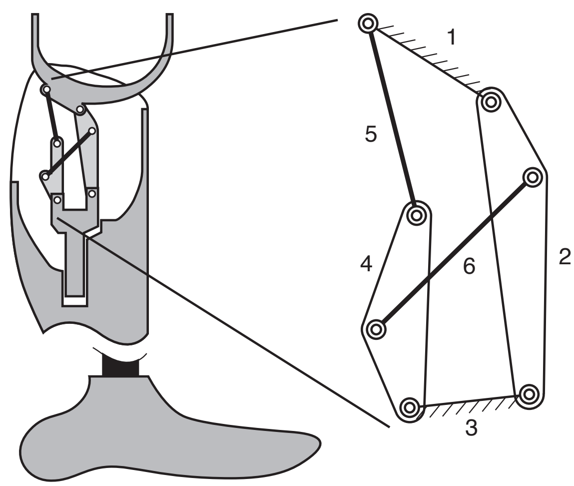


Figure 1.2 A 6-link prosthetic knee mechanism

Figure 1.3 shows an automobile convertible top mechanism with multiple links and joints. The mechanism is driven by a single actuator to guide the convertible top through a series of positions and orientations from a stowed position to a fully deployed position. This is a clever design of a mechanism that allows an intricate motion using a single actuator. Otherwise, numerous actuators would be required to guide the convertible top through the same successive positions. Such a design would be impractical due to excessive weight, cost and maintenance, to say nothing of a vastly more complex control scheme. In addition, a multiple actuator design would pose many additional challenges in meeting the packaging constraints within the tight space of the stowed position. Determining the right type of kinematic chain and sizing individual links to guide the convertible top precisely through successive positions can be a difficult but rewarding experience.

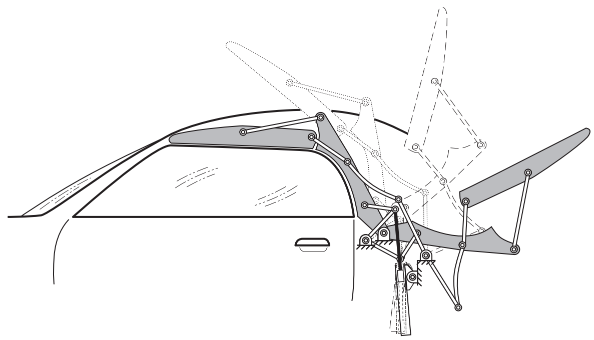


Figure 1.3 A convertible-top mechanism

Figure 1.4 shows a helicopter rotor blade mechanism that directs the blades to pitch back and forth while they are turning at a high speed. The advancing and the retreating blades pitch out of phase in each rotation. Rather than employing one heavy actuator on each fast moving blade, the task is accomplished by a robust, light weight mechanism. Here the mechanism is designed to transmit and control two different output motions; blade rotation and blade pitch. In kinematic terms we call it a differential with two kinematic degrees of freedom. There are many other examples of mechanisms that are designed to transmit and coordinate multiple motions and yet be driven by a single motor. The outputs in such devices are mechanically coupled to repeat the operations precisely and repeatedly without variation or failure.

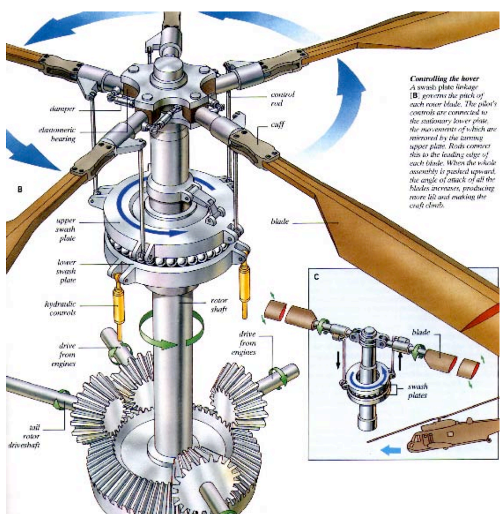


Figure 1.4 Helicopter rotor blade mechanism that transmits motion to the rotor shaft (rotation) and simultaneously allows the individual blades to pitch back and forth.

Figure 1.5 shows another example of a mechanism designed to produce multiple output motions from a single actuator. An important design improvement found in this orbital sabre saw is a mechanism that not only drives the blade up and down but also swivels the blade to avoid rubbing against the freshly cut surface during each return stroke.

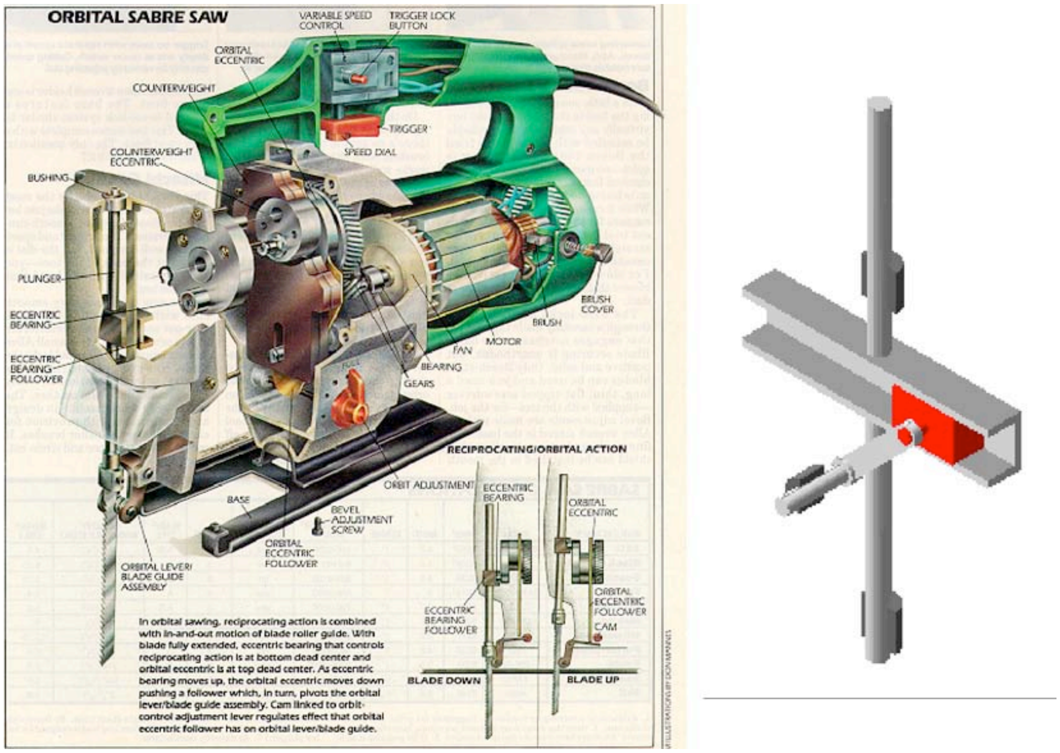


Figure 1.5 A four-link mechanism converts the rotary motion of the motor into reciprocating motion of the saw blade while the cam rotates and engages the blade during the upward cutting stroke.

Figure 1.6 shows a mechanism that draws a measured amount of wire from a spool and cuts it to a desired length several hundred times a minute. The motor drives the linkage, a four-link mechanism, whose output oscillates back and forth and is connected to friction rollers that grip the wire. As the output link oscillates back, a ratchet is employed so that only the forward motion of the oscillating output link is transmitted to the friction rollers; a cam attached to the motor shaft drives the blade, cutting the drawn wire to size. As indicated earlier, it is relatively easy to understand how such mechanisms work after they are designed. Conceptualizing the type of mechanisms, linkages, cams, ratchets, return springs etc. needed to carry out such tasks precisely can be both challenging and fun.

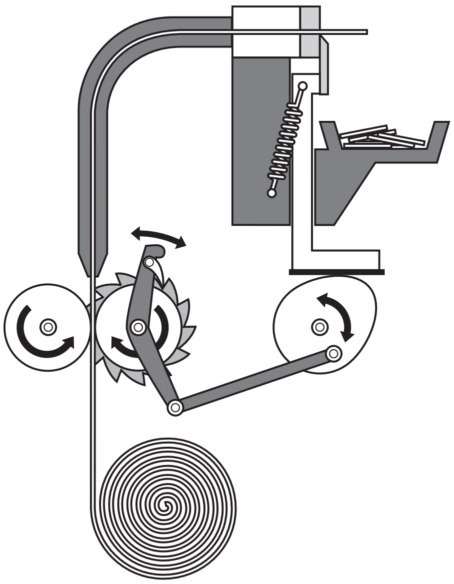


Figure 1.6 A wire-feeding and -cutting mechanism driven by a single motor. The four-link mechanism and the ratchet transmit motion to the friction rollers to draw the wire from a spool and the cam drives the blade to cut the wire to a pre-determined length.

Mechanisms are ubiquitous; from toys, surgical instruments, prosthetic devices, and landing gear, to devices for manufacturing and assembly automation. Some modern mechanisms are even too small to be seen by a naked eye, with link widths of less than a micron, thicknesses of a few microns and the link lengths less than 50 microns! The micro mechanism shown in Figure 1.7 is less than one-half square millimeter in size. This figure is an actual image of a system with two sets of electrostatic linear actuators placed 90 degrees to each other, and each imparting linear motion to a flexible mechanism which amplifies the motion 50 times. The amplified linear motion is then transmitted to the drive gear through a ratchet. The ratchet transmits only the forward motion to the gear. As the flexible mechanism retracts, without transmitting the reverse motion to the drive gear, the other actuator-mechanism assembly transmits a forward pulse to the gear thus keeping the gear turning continuously. The series of gears in the train convert rotational motion of the drive gear to the proper speed and torque.

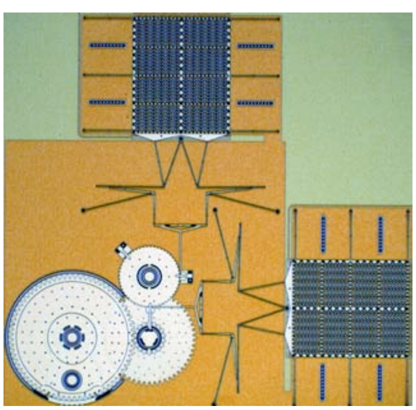


Figure 1.7 The micro engine shown here is driven by two electrostatic linear actuators each of which is connected through a motion amplifying compliant mechanism and a ratchet to drive the compound gear train continuously at high speed. The entire assembly fits within a 800 micron x 600 micron area.

1.1.1 KINEMATIC CHAINS

All of the mechanisms shown in Figures 1.1 – 1.7 are connected to a fixed frame through at least two joints. The links in these mechanisms form *closed kinematic chains* (see Figure 1.New (a)). A traditional robotic arm, like a human arm, is an *open kinematic chain* (see Figure 1.New (b)). The last link, the hand or the end effector is not connected to the frame or the body. In an open kinematic chain, the position of each link must then be actively controlled to keep the mechanism from collapsing. Thus, a six-degree of freedom robot must have six actuators, typically one at each of its joints, to control the motion of the end-effector. By comparison, a six-link closed-chain mechanism, such as the prosthetic knee of Figure 1.2, is controlled by only one. Actually, in the knee mechanism the motion of the thigh relative to the shank is the "actuation" and all the links follow through their respective angular displacements without collapsing when the single "input" is set in motion. All the mechanisms we deal with in this book are closed kinematic chains.

In an **open kinematic chain**, the final link is connected to the rest of the mechanism with only one joint.

In a **closed kinematic chain**, all of the links are connected to other links by at least two joints.

Open Versus Closed Kinematic Chains

Sometimes a given task can be performed by either a closed or an open kinematic chain. Consider the utility loader shown in Figure 1.8. One of the primary kinematic functions of the loader is to raise and lower the bucket straight up and down. The gross vertical motion can be accomplished in two fundamentally different ways; i.e., open or closed kinematic chains. Figure 1.8a shows an open kinematic chain where the individual link rotations are affected by two linear hydraulic actuators (pistons) on each side. Alternately, a four-link closed mechanism (Figure 1.8b) transmits the desired vertical straight-line motion to the bucket and is driven by a single actuator. The actuator is mounted to the frame thereby reducing the total inertia of the system. Furthermore since the linkage employs only pin-joints, it is an extremely reliable system requiring a minimal amount of maintenance. The bucket itself swivels for loading and unloading at any given position of the boom by a separate unrelated mechanism.

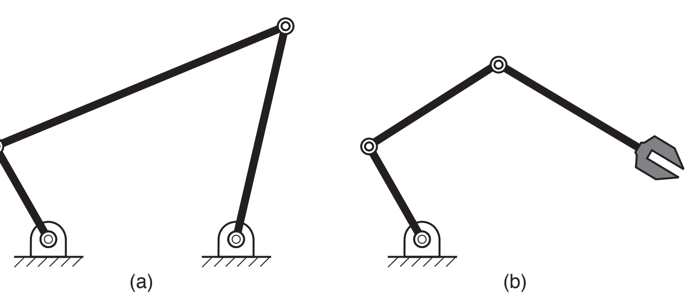


Figure 1. (a) A closed kinematic chain and (b) an open kinematic chain.

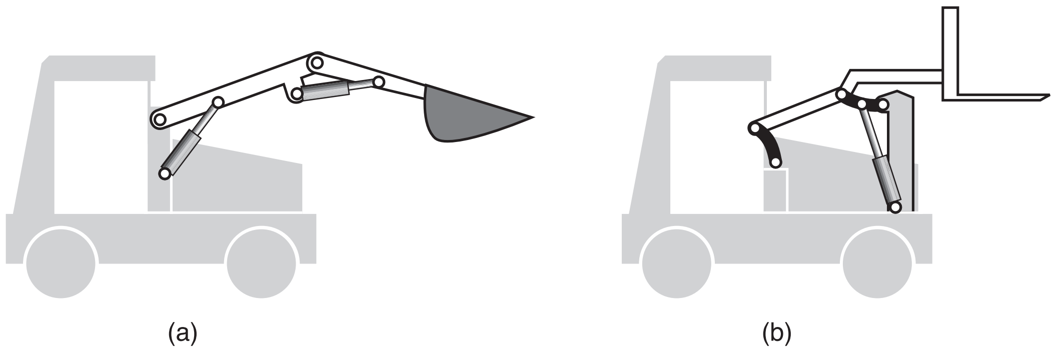


Figure 1.8 (a) A front-end loader design, an open-kinematic chain, with two actuators. (b) A four-link mechanism driven by a single actuator transmits approximately straight vertical motion to the bucket.

Closed kinematic chain mechanisms offer many positive advantages including high-speed capability, good repeatability, lower-cost, excellent reliability and minimal maintenance. On the other hand, open kinematic chains, with redundant actuators, can be reprogrammed to adapt to changing task requirements; that is, they are flexible. For instance, if the bucket on the front-end loader in Figure 1.8a needs to travel on a path that is different from straight vertical motion it can be readily reprogrammed. In a closed link chain, by contrast, a new mechanism with new linkage dimensions would have to be designed and installed for the mechanism shown in Figure 1.8b if a different path were required. But does the basic lift really have to travel on different paths or is a straight-line path sufficient? Be it construction equipment, manufacturing automation equipment, or consumer product design, the first issue to address is: how much flexibility (changeability) is truly needed. A summary of the characteristics for both closed and open kinematic chains is provided in Table New.

CHARACTERISTIC	CLOSED KINEMATIC CHAIN	OPEN KINEMATIC CHAIN
SPEED	Capable of high speed operation	Speed is limited by control and actuation
REPEATABILITY	Highly repeatable	Less repeatable
RELIABILITY	Typically very reliable with less maintenance required	More likely to require maintenance
FLEXIBILITY	Little flexibility in range of tasks performed	Very flexible, easily reprogrammed for new tasks
COST	Lower cost	Higher cost

Table 1. Summary of characteristics for closed and open kinematic chains.

In general, closed kinematic chain mechanisms are preferred for

a) rigid body guidance when the path of at least one point on a link and relative angles of rotations are highly defined. Examples include an automotive suspension mechanism that guides the wheel rectilinearly.

b) non-linear oscillation or reciprocating motion where one or more such output motions are to be driven by a single input.

A system has **redundant actuators** if it has more than the minimum number of actuators required to accomplish the intended motion.

Designers typically resort to use of redundant actuators only if the system is required to adapt to *unplanned* operational requirements. However, neither poor planning during early phases of design/procurement, nor insufficient time to design a mechanism, can justify the use of systems with redundant actuators. The initial investment and the cost of repair and maintenance should always be taken into strong consideration. The following scenarios are presented to highlight why closed kinematic chain mechanisms are preferred over redundant-actuator based designs when considering cost, reliability, efficiency, and weight considerations.

Closed Kinematic Chains:

1. When a prime mover (e.g., a motor or other actuator) already exists in the system, a mechanism can be designed to tap some of the power to accomplish a secondary function. A simple example is the mechanism of a lawn sprinkler (see Figure 1.9) which uses the kinetic energy of the water to oscillate the sprinkler back and forth. Another example of a four-bar linkage is used in the Roller blade ABT brake system shown in Figure 1.10. In this case, the boot cuff, which is firmly clamped to the lower leg, acts as the prime mover. When braking is desired, the in-line skater moves his/her toe forward, causing rotation of the lower leg about the ankle joint. The resulting relative rotation between the cuff and the boot moves the brake pad down into contact with the skating surface. This mechanism is also adjustable; the coupler link is designed to lengthen and adjust the response of the linkage as the brake pad wears down. Of course, the same task can be accomplished by two sets of actuators (say solenoids) per boot with position or pressure sensors mounted on each cuff. Another alternate might be a cable-actuated (by hand) cylinder. The linkage design of Figure 1.10 is, however, simple, reliable and cost-effective.

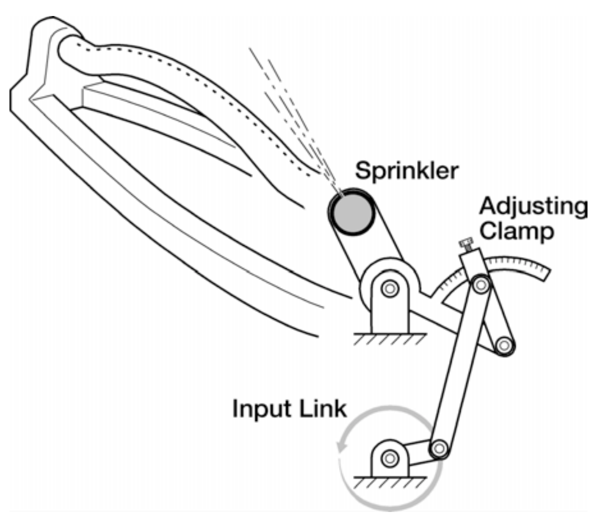


Figure 1.9 An adjustable lawn-sprinkler mechanism propelled by the water pressure applied to rotate link A_o-A.

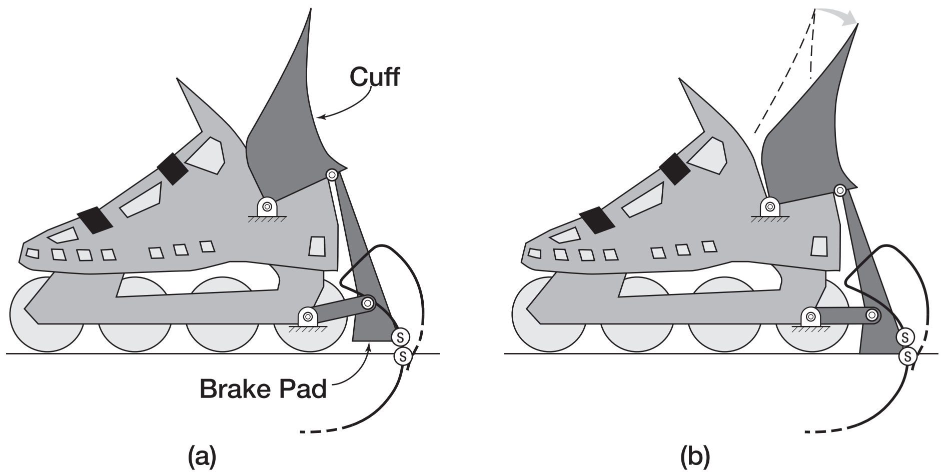
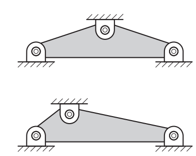


Figure 1.10 Roller blade ABT brake system. in which the cuff-link provides the input to guide the brake pad.

A **whiffletree** linkage has a link pinned at a point between its two ends, with additional connections on the ends of the link.



2. When a rigid body needs to be guided precisely through a repeated set of desired positions and orientations. Examples include an automobile suspension mechanism, a knee brace, an automobile convertible top linkage (as shown in Figure 1.3), and an aircraft landing gear.

3. When a single input is used to control multiple outputs motions. A cost-effective solution would be to use simple linkages in parallel. Examples include multiple leaves of a shutter in a camera operated by a single input as well as the orbital saw and helicopter rotor blade mechanisms discussed previously.

4. When it is desired to (a) control one or more outputs with a single input or (b) transfer the power from two or more independent inputs to control the motion of a single output a two (or more) degrees of freedom "differential" mechanism can be employed. The principle of a mechanical differential can be realized with gears (bevel or epicyclic), belts and pulleys, or a linkage arranged in a whiffletree fashion.

5. When the desired motion is a simple rotation about a fixed axis or a translation along a fixed axis, with a non-linear relation between the input and the output motions a cam (or a linkage if space permits) can be employed. Examples include an automotive valve train or any cam-follower system.

6. When high speed motion or manipulation is needed as in high speed manufacturing or assembly operations. For instance, if the production rates are over 200 parts per minute, the need to reduce the mass and therefore the inertia is very great. Robots with a series of cantilevers connected by massive actuators would be of little help. A mechanism driven by a single actuator connected to a fixed frame can perform at high speeds with excellent repeatability and reliability. Cam-driven systems are particularly suited for such applications.

7. When mechanism timing is critical. A closed kinematic chain with a single actuator ensures that the various motions will always have the proper sequence and timing. For example, in the feed and cut application in Figure 1.6 the feed and cut operations must be properly timed. Another example

would be a sewing machine. Mechanically tying the various motions together via closed kinematic chains ensures that the mechanism will function properly with the upper and lower threads properly coordinated to form the stitches.

1.2 PLANAR AND SPATIAL MOTION

> **Planar rigid-body motion** consists of rotation about an axis perpendicular to the plane of motion and translation within that plane.

A large majority of mechanisms undergo motions in which all the links move in parallel planes. This section emphasizes this type of motion, which is called *co-planar* or simply *planar motion*. Planar rigid-body motion consists of rotation about an axis perpendicular to the plane of motion, and translation within that plane. Translation means that all points in the body move along parallel straight or planar curvilinear paths and all lines embedded in the body remain parallel to their original orientation. Figure 1.11 shows an automobile hood linkage in which all the links move in parallel planes. In these discussions, all links are assumed to be rigid bodies.

Spatial mechanisms allow movement in three dimensions. Figure 1.12 shows a spatial mechanism that converts a rotary motion into reciprocating motion. This is called a Swash-plate mechanism and is used in pumps. Of course the same mechanism can be used to convert a reciprocating motion input into a rotary output motion. Certain types of Stirling engines employ such a mechanism. The swash plate does not move parallel to a single plane. A point on the surface of the swash plate traces a three-dimensional trajectory during one cycle.

In some cases, a spatial mechanism can be decomposed into a series of planar mechanisms with one or perhaps two links that do not move in a single plane. One mechanism might move vertically with another in a horizontal plane while a single link could connect the two.

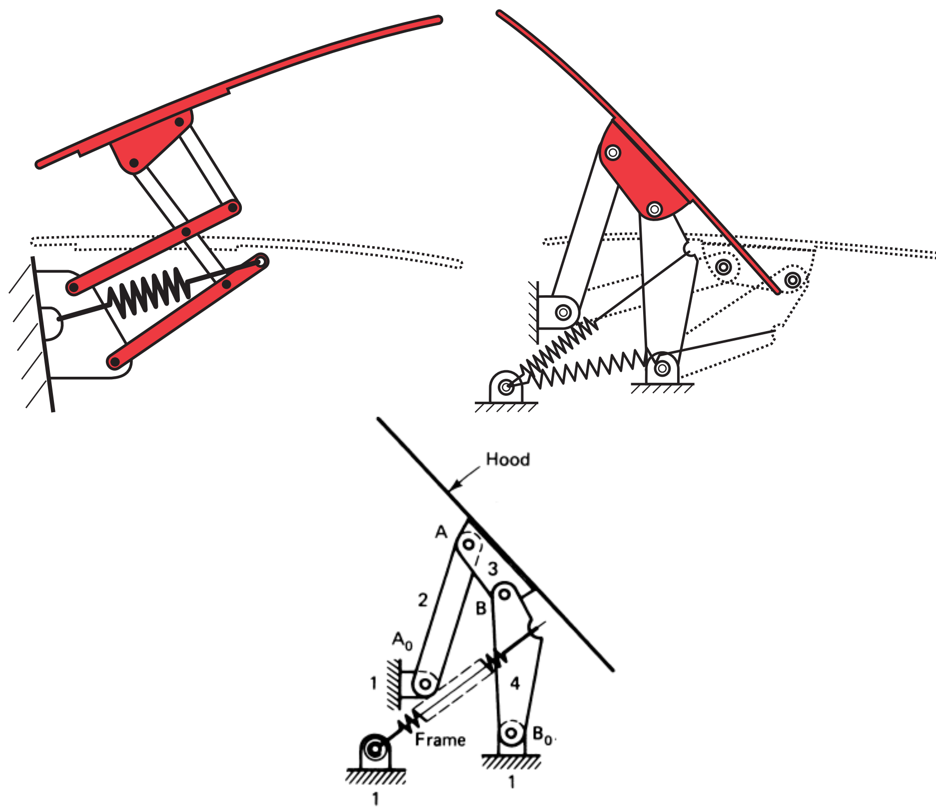


Figure 1.11 Automobile hood mechanisms as examples of a planar mechanism.
(a) A six link mechanism and (b) four-link mechanism.

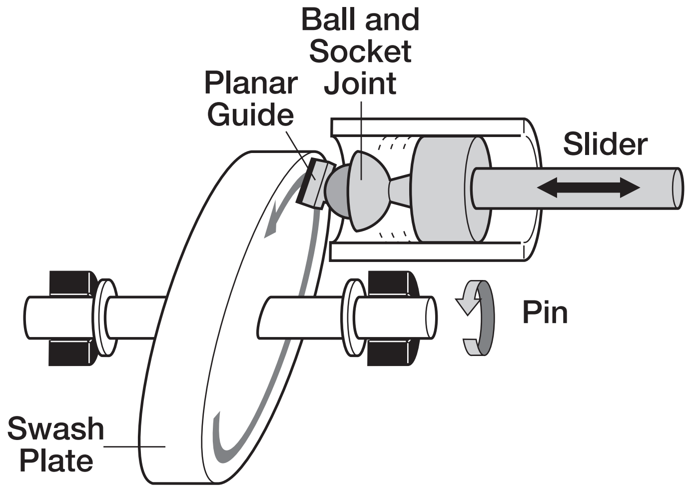


Figure 1.12 A 3-D swash plate mechanism that converts rotary motion into reciprocating motion.

Not all joints allow the same number and types of degrees of freedom. Recognizing the types of joints, and their characteristics, is critical in the analysis and design of mechanisms.

1.3 JOINTS AND LINKS

The gross motion of an end-effector attached to an output link of a mechanism is affected by the number and types of both the links and joints in the mechanism. Let us first consider the joints. Table 1.1 shows different types of kinematic joints, the manner in which they are represented in a kinematic diagram, and the number of degrees of freedom that each allow. The hood mechanism shown in Figure 1.11 employs only hinge or pin joints. The swash plate mechanism shown in Figure 1.12 has a pin joint (bearing) between the shaft and the frame, a *ball-and-socket joint*, a sliding joint between the piston and the cylinder and a *plane joint*. Different types of joints allow different number of degrees of freedom and different types of degrees of kinematic freedom. A pin joint between two rigid bodies allows a single rotational degree of freedom. The two connected bodies can only rotate with respect to each other. A round piston inside a cylinder can slide inside the cylinder and can also rotate about its common longitudinal axis.

Gross motion: the path and orientation of a body resulting from all of the influences on it.

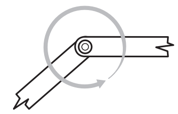
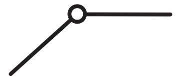
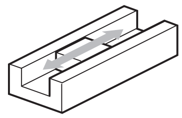
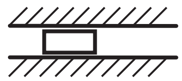
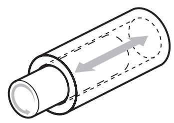
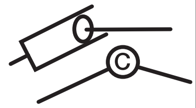
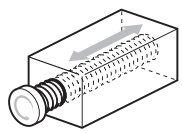
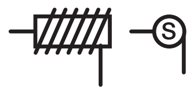
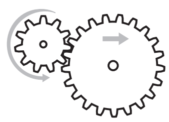
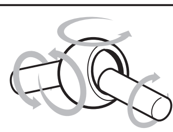
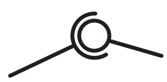
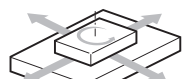
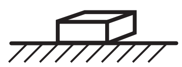
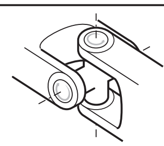
Joint Type	Depiction	Schematic	Example	Degrees of Freedom
Pin Revolute				1 Rotational
Slider Prismatic				1 Translational
Slider Cylindrical				2 1-Rotational 1-Translational
Screw				1 1-Rotational or 1-Translational (not independent motions)
Gear				2 1-Rotational 1-Translational
Ball-and-Socket				3 3-Rotational (R_x, R_y, R_z)
Planar				3 1-Rotational (R_z) 1-Translational (T_x, T_y)
Universal				2 Rotational

Table 1.1 Types of joints and their kinematic degrees of freedom

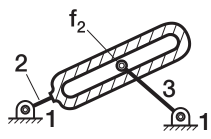
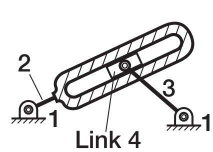
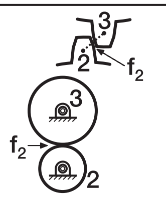
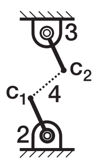
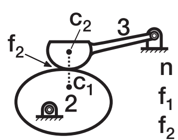
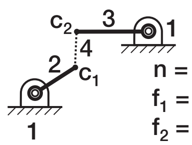
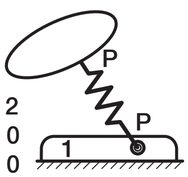
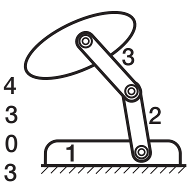
Joint Type	Higher Pair	Lower Pair Equivalent	Remarks
Pin-in-Slot			Replace f_2 joint with two f_1 joints and a binary link 4
Gear	 $n = 3$ $f_1 = 2$ $f_2 = 1$ $F = 1$	 $n = 4$ $f_1 = 4$ $f_2 = 0$ $F = 1$	Replace the higher pair joint between the meshing gears with a link 4 and two f_1 joints, c_1 and c_2. Locations of c_1 and c_2 are the centers of curvature of respective gear profiles at that instant.
Cam	 $n = 3$ $f_1 = 2$ $f_2 = 1$ $F = 1$	 $n = 4$ $f_1 = 4$ $f_2 = 0$ $F = 1$	Replace the higher pair joint between the cam and the follower with link 4 and f_1 joints at c_1 and c_2 (located at the centers of osculating circles).
Spring	$n = 2$ $f_1 = 0$ $f_2 = 0$ 	$n = 4$ $f_1 = 3$ $f_2 = 0$ $F = 3$ 	Replace a spring connecting two rigid bodies with a dyad-two rigid links connected by an f_1 joint each connected to the rigid bodies by f_1 joints (total of three f_1 joints)

Table 1.2 Higher-pair joints and their instantaneous lower-pair equivalents

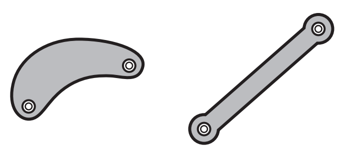
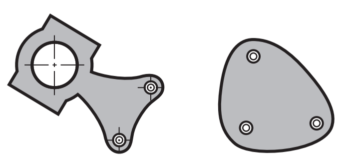
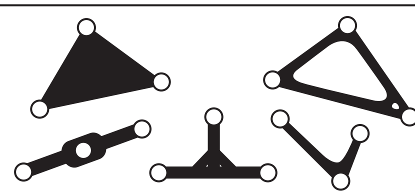
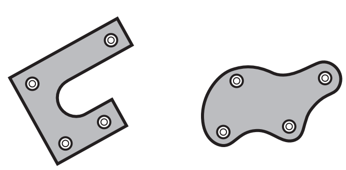
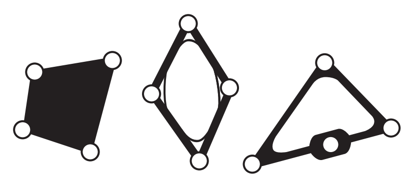
Link Type	Typical Form	Skeleton Diagrams
Binary		
Ternary		
Quaternary		

Table 1.3 Link types

Such a joint, called a cylindrical joint, allows two degrees of freedom. An example is the ink cartridge inside the barrel or cap of a ballpoint pen. If instead the piston is changed to a rectangular cross-section, and the "cylinder" is simply a gap between two plates (or a slot cut in a plate), then the piston can only slide and not rotate. That is a *prismatic joint* with one (sliding) degree of freedom. A ball-and-socket joint allows three rotational degrees of freedom. A plane joint, like an eraser on a blackboard, allows rotation about the axis perpendicular to the plane (blackboard) and translation anywhere within the two dimensional plane. A screw and a nut assembly allows one degree of freedom, screw thread pitch determines how far the nut translates when the screw is turned 360 degrees.

Another way to think about the function of a joint in connecting rigid bodies is that it constrains them to certain type of motion. For instance, a pin joint constrains the connecting links to only have rotational motion with respect to each other. We will return to this line of reasoning later in our computation of total degrees of freedom of a mechanism.

Selection of joint type typically depends on the motion characteristics desired in a mechanism. But it is also important that the mechanism be easy to maintain, and last a long time. Joints with surface contact distribute loads better than the ones with line contact. For instance, pin joints have surface contacts (sleeve and journal) and are easy to maintain. Sliding joints too have surface contact but are susceptible to increased friction due to debris buildup, inaccuracies due to wear and in extreme cases may tend to jam. Cam and follower joints have a line contact and are difficult to maintain since the line of contact tends to skim any lubricant out of the joint. Cost is also an issue; pins are inexpensive, standard components that can be quickly obtained from a variety of sources. Joints with surface contact are called **lower pairs** and the joints with line or point contact are called **higher-pair** joints. Higher pair joints should be used for certain benefits they offer. For example, using higher-pair joints may allow you to design a more compact mechanism which can be very important in some applications where physical space is at a premium. They typically have fewer components as well, which can simplify the assembly of the mechanism. Table 1.2 shows lower-pair equivalent mechanisms of higher-pair joints. In this table, joints are labeled f_1 and f_2, referring to the number of degrees of freedom *allowed* by the joint. The number of these joints combined with the number of rigid bodies (n) are used to calculate the total degrees of freedom (F), as will be discussed in Section 1.5.

In a closed kinematic chain mechanism, the fixed link must have at least two joints, each connecting two other links. A link with two joints is called a *binary link*, the one with three is called a *ternary link*, and the link with four joints is called a *quaternary link*. Table 1.3 shows different link types and their representation in a skeleton diagram. Planar mechanisms with more than one kinematic chain, called multi-loop mechanisms, will often have a combination of link types. For example, the motorcycle suspension of Figure 1.1 has four binary links (links 1, 3, 5 and 6) and two ternary links (links 2 and 4). Notice also that all connections are pin joints except the slider joint between links 5 and 6.

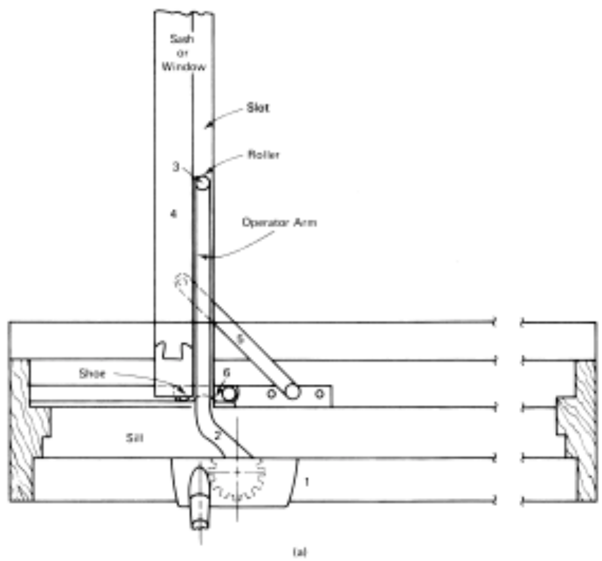
Kinematic Diagram: a simple schematic showing the relative positions and types of connections of the links in a mechanism. **Links** are designated with Numbers, **Joints** with letters.

1.4 KINEMATIC DIAGRAMS

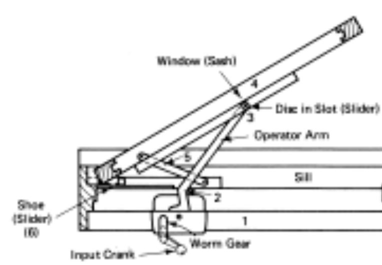
It is often difficult to visualize the movement of a multi-loop mechanism such as that shown in Figure 1.13a, especially when other components appear in the same diagram. The first step in the motion analysis of complicated mechanisms is to sketch a simple *skeleton* or "kinematic" diagram. This requires sketching a "stripped-down" stick diagram, such as that shown in Figure 1.13c. A kinematic diagram clarifies how various links are interconnected and the types of joints used. It masks the actual shapes of the links since link shape does not necessarily affect the essential kinematic behavior or function of the mechanism. The skeleton diagram serves a purpose similar to that of an electrical schematic or circuit diagram in that it displays only the essential skeleton of the mechanism. Before we begin any serious analysis of a mechanism, we must clearly understand relationships between the number and types of links and joints that make up the mechanism.

Figure 1.13 A sectional top-view of a casement window mechanism is shown in its open position in (a) and nearly closed position in (b). The kinematic diagram shown in (c) illustrating how various links are interconnected.

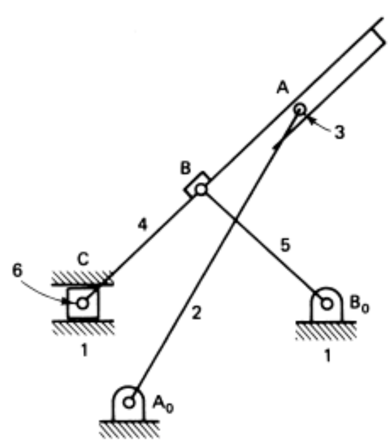
(a)



(b)



(c)



All links must be numbered in a kinematic diagram typically starting with the ground link as number 1. It is also recommended that joints be lettered. Figure 1.13 shows a casement window mechanism whose purpose is to operate the opening and closing of a residential window from the inside of the house by rotating an input crank. Note also that, because of this design, in the open position, both sides of the window can be cleaned from the inside by access on the left side of the mechanism. Figure 1.13c shows the kinematic diagram (sketch) for the casement window linkage. Notice that there are six links, five pin joints, one slider joint, and one roller in this sketch. Links 1 and 4 are ternary links. The kinematic diagram simplifies the mechanism for visual inspection and, if drawn to scale, provides the means for further analysis.

1.5 DEGREES OF FREEDOM

Consider the chain of binary links connected by pin joints as shown in Figure 1.14. If a motor is attached to the input link 2, the output link 6 may not respond directly – there seems to be too many intervening links! Clearly, there is a need for some rules of mobility by which mechanisms can be put together. A closed chain mechanism must have at least three links. With one link fixed, a three-link mechanism has two moving links.

Degrees of freedom: the number of independent variables that must be specified in order to fully define the position and orientation of a body.

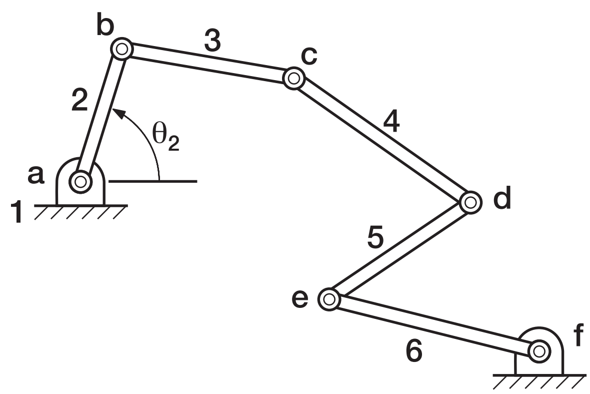


Figure 1.14 A closed kinematic chain consisting of 6 binary links

If all three links are connected to each other by a single degree of freedom joint, say pin joints, the assembly behaves like a truss structure (see Figure 1.15a). However if we replace the pin joint between link 2 and 3 with a two degree of freedom joint, then the assembly behaves like a mechanism. The familiar pair of spur gears shown in Figure 1.15b illustrates this point since the joint between the meshing gears allows both relative rotation and sliding; it is a two degree-of-freedom joint.

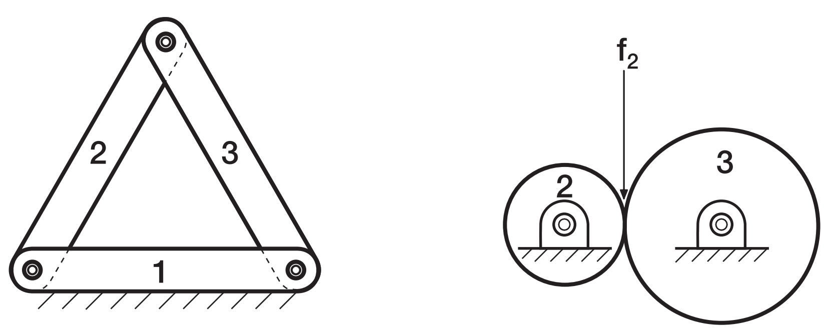


Figure 1.15 (a) A truss *structure* with 3 links. (b) A mechanism with 3 links and an f_2 joint .

Now suppose, for the moment, we limit ourselves to only "f_1" type joints referred to here as pin joints. What is the minimum number of links one must have in a closed kinematic chain to have mobility. Since we know that a three link closed chain yields a structure, let's try four links. Four links, to be precise four binary links, connected to each other by pin joints yields a four-bar mechanism as shown in Figure 1.16. It seems intuitive that if a motor is attached to link 2 (to rotate link 2 relative to link 1 to angle θ_2), the output link 4 will respond directly. When we say 'respond directly' we implicitly mean that if we know or fix the position of just one link, (in this case the input link 2), the position and orientation of all other links including link 4 is defined. So the angular position of link 2 is the only independent variable needed to define the kinematic mobility of the four-bar mechanism. Now let us uncover that fundamental rule of mobility for any planar mechanism.

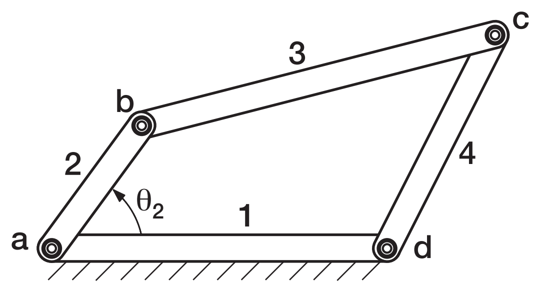


Figure 1.16 A four-bar mechanism

An **Unconstrained Rigid Link** in a plane has three degrees of freedom. Each connection between links reduces the total number of degrees of freedom by 1.

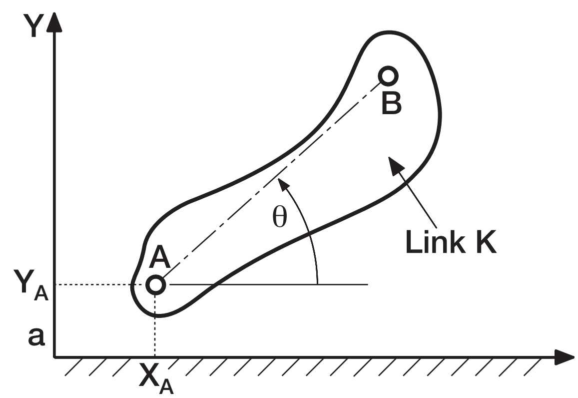


Figure 1.17 A rigid link rotating freely about a fixed axis.

Suppose that the exact position of rigid link K is required in a *planar coordinate system* (XY) as depicted in Figure 1.17. How many independent variables will completely specify the position of this link? The location of point A can be reached, say, from the origin by first moving along the X axis by x_A and y_A in the direction of the Y axis. Thus, these two coordinates, representing two translations, locate point A. More information is required, however, to completely define the position of link K. If the angle θ of the line connecting points A and B with respect to the X axis is known, the position of link K is specified in the plane XY. Thus there are *three independent variables*: x_A, y_A, and θ (two translations and one rotation, or three independent coordinates) associated with the position of a link in the plane. In other words, an unconstrained rigid link in the plane has three degrees of freedom.

If there is a conglomeration of *n* separate links, they possess a total of 3*n* degrees of freedom before they are joined to form a linkage system. As the mechanism is built up of separate links, each fixation or linkage eliminates a degree of freedom. Since one link must be designated as the fixed link in a mechanism, a conglomeration of n-links has $3n$ -3 degrees of freedom. That is, (n-1) moving links possess 3(n-1) degrees of freedom. Connections between links result in further loss of total degrees of freedom of the system of links. As we know, a pin (revolute) joint only allows relative rotation between the connecting links and hence it *removes two degrees of freedom*. If point A on the link in Figure 1.17 is a pin joint between link K and ground, then two independent variables, x_A and y_A, are now fixed, leaving θ as the one remaining degree of freedom of link K.

In an assembly of links such as that in Figure 1.14, each pin connection will remove two degrees of freedom of relative motion between successive links. This observation suggests that the *degrees of freedom, F,* of an n-link mechanism connected only by f_1 type (pin or sliding) joints, is

$$F = 3(n - 1) - 2f_1 \tag{1.1}$$

The other type of joint that can exist in a planar mechanism is an f_2 joint, or one that allows two degrees of freedom. Each f_2 type joint in a mechanism removes one degree of freedom. Therefore, the degrees of freedom of a planar mechanism with f_1 and f_2 type joints is given by

$$F = 3(n - 1) - 2f_1 - f_2 \tag{1.2}$$

An example of a mechanism with an f_2 joint is the set of spur gears shown in Figure 1.15b. The gear contact is an f_2 joint so that

$$F = 3(3 - 1) - 2(2) - 1(1) \tag{1.3}$$

$$= 6 - 4 - 1 = +1$$

yielding one degree of freedom for the gear pair.

Table 1.2 shows conversion of f_2 joints to f_1 joints. Simply stated, an f_2 joint is equivalent to two f_1 joints and an additional link in the mechanism as shown in Figure 1.18. Notice that the rotational and translational degrees of freedom of the fork joint in (a) have been directly replaced by the rotation of the pin joint between links 2 and 4 and the translation by the slider between links 4 and 3.

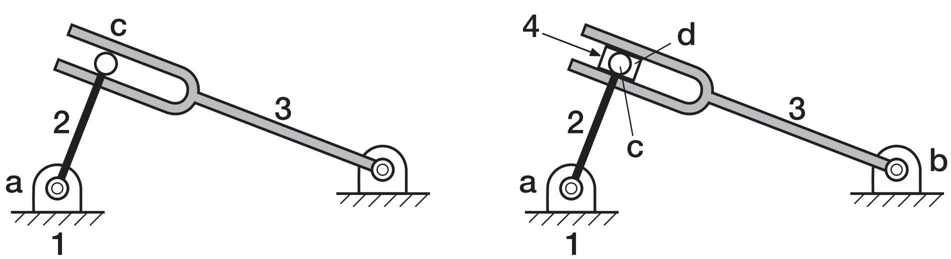


Figure 1.18 Replacing a pin-in-slot higher pair joint (fork joint) with a lower-pair equivalent

Equation 1.2 is called the Gruebler's Equation for determining degrees of freedom of a planar mechanism. There are other forms of mobility equations in kinematics literature, but we shall not discuss them in this chapter. For a spatial mechanism, the three-dimensional version of Gruebler's equation takes the form of

$$F = 6(n - 1) - 5f_1 - 4 f_2 - 3f_3 - 2f_4 - f_5 \tag{1.4}$$

An unconstrained rigid link that is free to move in a three-dimensional space has six degrees of freedom – motion along x, y and z axes and rotations about x (pitch), y (roll) and z (yaw) axis. Since joints that allow four and five degrees of freedom are not often employed, we normally restrict the number of terms to

$$F = 6(n-1) - 5f_1 - 4f_2 - 3f_3 \tag{1.5}$$

Returning to the planar version of Gruebler's equation, let us re-consider the four-bar linkage shown in Figure 1.16. Since all four joints are *f1* type (pin) joints, we use *f1* = 4, *n* = 4 and $f_2 = 0$ in Gruebler's equation which yields the total number of kinematic degrees of freedom of the mechanism

$$F = 3(4-1) - 2(4) - 0 = +1$$

A structure with no mobility has $F = 0$. For instance, the truss structure shown in Figure 1.15a has n = 3, $f_1 = 3$ and $f_2 = 0$ yielding $F = 3(3-1)\ 2(3) - 0 = 0$.

Mechanisms with two degrees of freedom require two independent inputs to define and control the motion of all other links in the mechanism. Mechanisms with one degree of freedom require only a single actuator.

Except in some special cases which we will discuss later, Gruebler's equation can be used to determine the degrees of freedom of mechanisms. However, it is recommended that students verify the computed degrees of freedom intuitively as well. This will not only help gain insight to the kinematics of motion, but will also help detect any inadvertent errors in the identification of joints. The joint-weld method is one such approach.

1.5.1 JOINT-WELD METHOD

The joint-weld method is an intuitive method to verify the degrees of freedom of a mechanism. Consider the four-bar linkage shown in Figure 1.16. We already know that this mechanism has one degree of freedom implying that we need to specify only one variable (say the joint angle θ_2) to determine the positions and orientations of all links in the mechanism. Another way to interpret this is to say that if we fix one variable, that is freeze the joint (variable θ_2), the entire mechanism is fixed. So then, we simply freeze (weld the jointed connection) one joint variable at a time until the mechanism has zero mobility (becomes a structure) and equate the number of joint-welds to degrees of freedom. In the case of the four-bar mechanism of Figure 1.19a, it should take only one such joint-weld. If we weld the jointed connection between links 1 and 2 by welding joint a, link 2 becomes fixed in that position, leaving only two moving links, 3 and 4. The loop b-c-d becomes a truss structure and has zero mobility. Welding one joint turns the mechanism into a structure and therefore the mechanism has one degree of freedom.

Now let us consider the casement window mechanism shown in Figure 1.19b. First, we convert all higher pairs to lower-pair equivalents so that each weld removes only one degree of freedom. Figure 1.19b(i) and (ii) show that the f_2 pin-in-slot joint A is decomposed into an extra link and two f_1 joints, a pin joint and a sliding joint. If we weld the sliding component of joint C (see Figure 1.19b(iii) and (iv)), the loop C-B-B_o turns into a truss structure thereby fixing the positions of links 4 and 5 (Figure 1.19b(v)). With link 4 fixed in its position, link 2 also loses its mobility. The entire mechanism becomes an immobile structure by simply welding the slider. Hence the mechanism has one degree of freedom.

Let us try this method on the six-link mechanism shown in Figure 1.19c. By welding joint a, we fix the position of link 2. The loop b-c-d-e-f still has mobility. Welding joint b fixes link 3 leaving loop c-d-e-f with one degree of freedom.

Further welding joint c, we fix link 4 and the loop d-e-f forms a truss structure. That is, links 1, 6, and 5 form a closed-loop truss structure. We had to weld three jointed connections, a, b, and c to turn the mechanism into a truss structure; therefore the mechanism has three degrees of freedom. Note that all the three joints we welded are single degree of freedom joints. We essentially add the degrees of freedom of all the joints we weld.

The procedure for the joint-weld method can be summarized as:

a) Convert all higher pair joints to lower pair equivalents.
b) Weld the jointed connection (lower-pair joint) between a moving link and the ground link.
c) The moving link is now fixed. All other joints on this link can now be represented as "ground joints."
d) Starting with each new "ground joint" on the recently fixed moving link count the joints to the next ground joint. If the loop has only three f_1 type joints, then it is a truss structure. If not, weld the next joint in the loop.

Once you practice this method on a few familiar mechanisms, you will learn to readily recognize zero degree of freedom truss structure loops and single degree of freedom four-bar loops. You will find this method very useful, especially in understanding the kinematic nature of otherwise seemingly complex mechanisms and determining their degrees of freedom.

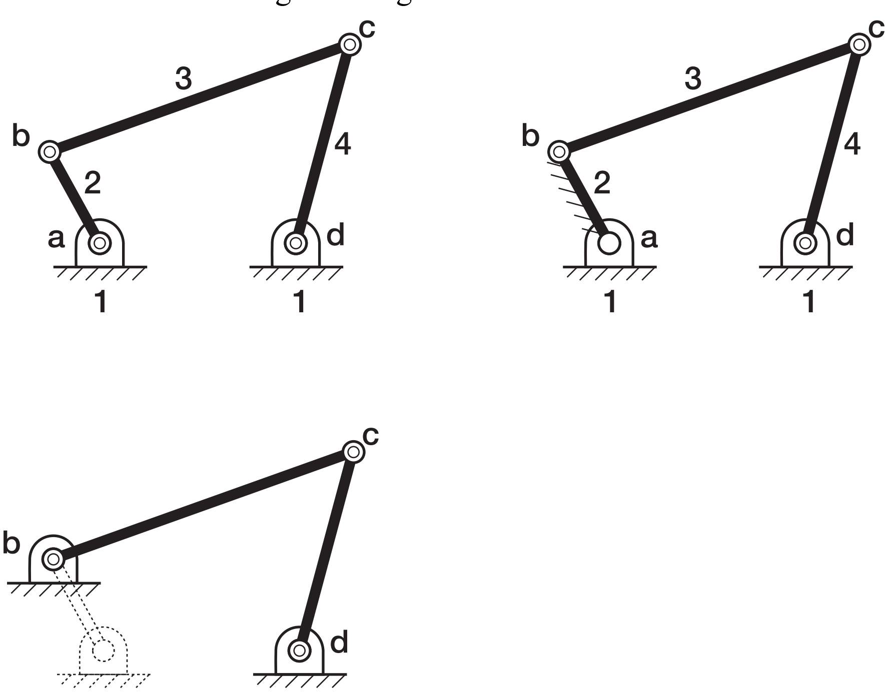


Figure 1.19(a)

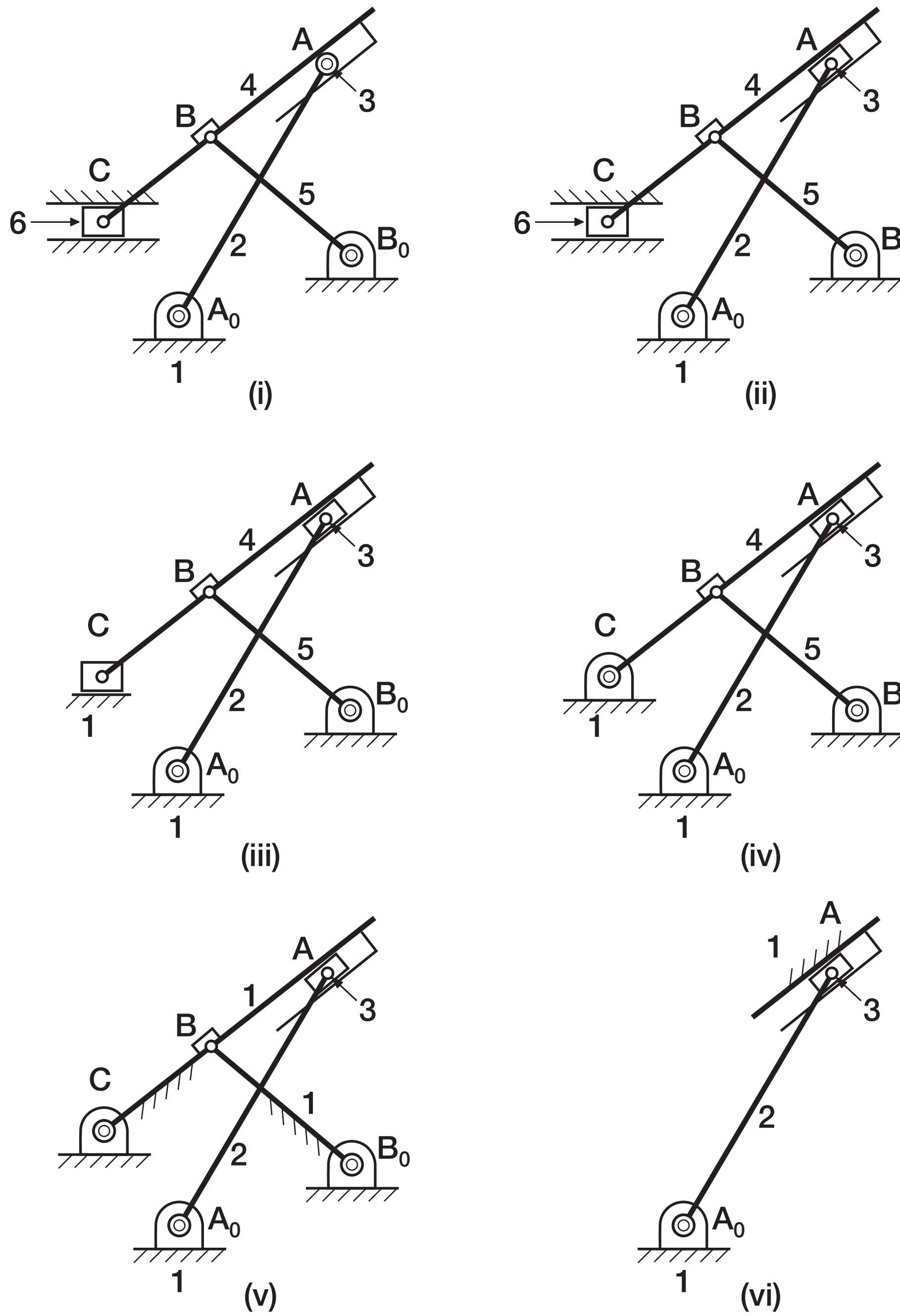


Figure 1.19 (b)

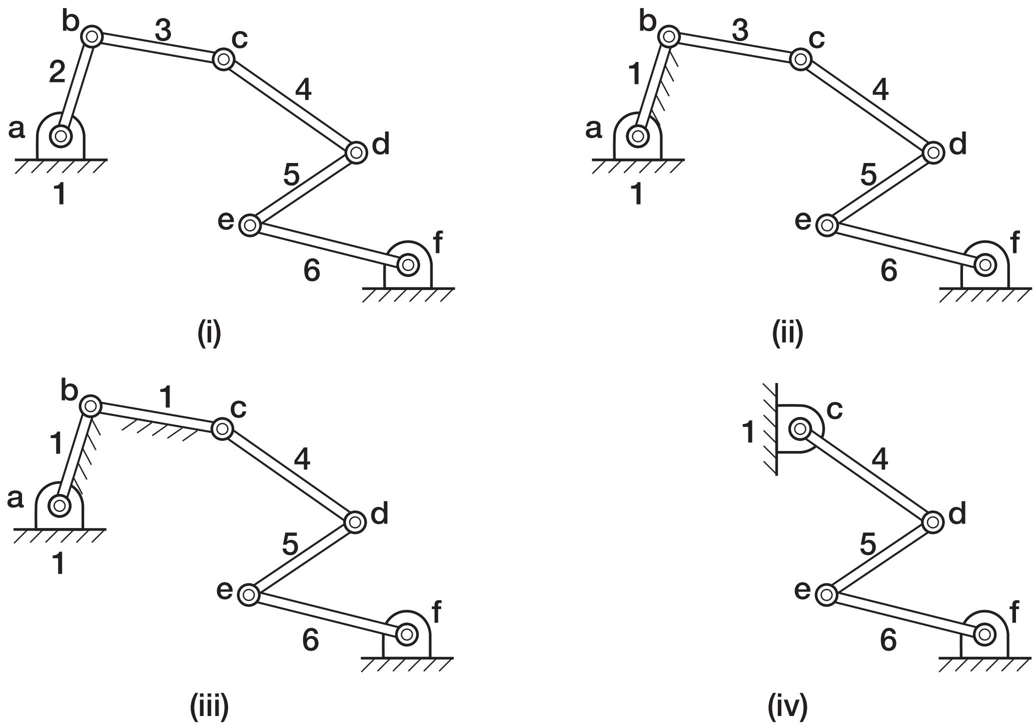


Figure 1.19 (c)

Gruebler's Equation can lead to incorrect results. Special geometric cases may allow more degrees of freedom than Gruebler's equation indicates. In the case of trivial degrees of freedom, Gruebler's equation may indicate that there are more degrees of freedom than are actually present.

1.5.2 SPECIAL CASES OF GRUEBLER'S EQUATION

It should be noted that there are mechanisms whose computed degrees of freedom, using Gruebler's equation, may be zero indicating that the device is a structure or may be negative indicating an indeterminate structure. Some of these devices can, nevertheless, move due to special proportions and/or symmetry. As an example, for the five-bar linkage shown in Figure 1.20, $F = 3(5 - 1) - 2(6) = 0$. However, the linkage moves because of its *parallelogram* configuration. This is an over-constrained linkage in which the third grounded link provides a redundant constraint. Like any over-constrained design, this linkage is very sensitive to manufacturing errors. That is, the linkage will jam if links 2, 3, and 4 are not exactly the same length, or if there is even a slight difference in the distance between pin connections on links 1 and 5.

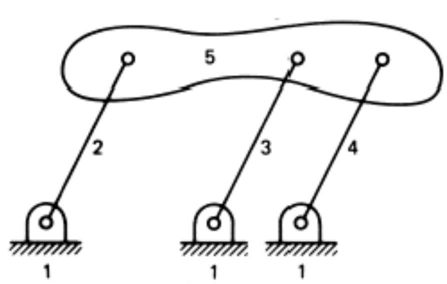


Figure 1.20 An over-constrained mechanism with mobility due to its special configuration.

Figure 1.21 shows another over-constrained example. Here we see two nip rollers in "pure" (zero tolerance) rolling contact. It can be shown that pure rolling contact allows only one degree of freedom, making it an f_1 joint that removes two degrees of freedom. Gruebler's equation yields F = 3(3 – 1) – 2(3) – 0 = 0. This simple mechanism does move, due simply to the fact that the sum of the radii of the rollers equals the distance between the ground pivots for all positions of the rollers. Assuming the rollers are rigid, even slight over-sizing of the radii would make the mechanism impossible to assemble. Under-sizing the radii would result in no rolling at all.

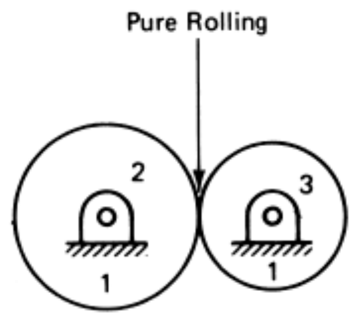


Figure 1.21 Two nip rollers in pure rolling contact.

There are also instances when Gruebler's formula yields an excessive number of degrees of freedom. This may involve a passive or redundant degree of freedom and does not alter the constraint between the specified output versus input motions of a mechanism. Take, for example, the cam-and-follower mechanism of Figure 1.22. Here the rotation of the follower roller 4 does not affect the output oscillation of the follower arm 3. Even if roller 4 were welded to arm 3, the motion of arm 3 would remain unchanged. Checking this condition by Gruebler's equation, regarding the cam and roller contact as a joint that allows both rolling and sliding (f_2 joint),

$$F = 3(4 - 1) - 2(3) - 1(1) = +2$$

Now, if we weld roller 4 to arm 3,

$$F = 3(3 - 1) - 2(2) - 1(1) = +1$$

Also, if slipping (the redundant freedom between cam and roller) is prevented by friction—in other words, their contact becomes rolling only —

$$F = 3(4 - 1) - 2(4) = +1$$

The reader can verify that the casement window mechanism in Figure 1.13 contains a similar passive degree of freedom if the roller (3) is allowed to slip in the slot in link 4.

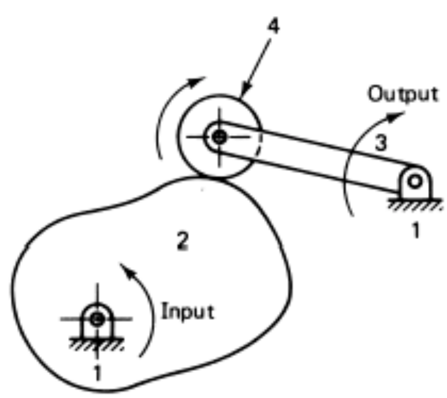


Figure 1.22 A cam and an oscillating roller follower

1.6 COMMON CLOSED-CHAIN MULTI-LINK MECHANISMS

In this section we will look at some mechanisms categorized by the number of links in their closed kinematic chain.

1.6.1 FOUR-BAR MECHANISMS: SIMPLE AND VERSATILE

A four -bar linkage with all f_1 type joints is the simplest single-degree of freedom mechanism and is used in many applications. Figure 1.23 shows a four-bar mechanism used by 3M Corporation for picking flat disk covers out of a magazine and place them one-at-a-time on a moving conveyor – many times per minute. The linkage is designed so that the *floating link* (or the *coupler link*) orients itself precisely with the part in the pick-up position and gently places the product horizontally on the conveyor. One of the links is attached to the ground and is driven by an actuator so that the end-effector performs the pick-and-place operation at high speeds. Additional demands placed on the linkage designer may include effective motion transmission, matching the speed of the coupler with that of the conveyor at the moment the part is released on the conveyor. Note that a gripper is rigidly connected to the coupler link that grasps and releases the part. The exact grasp and release time can be coordinated with the input crank angle either mechanically or electronically.

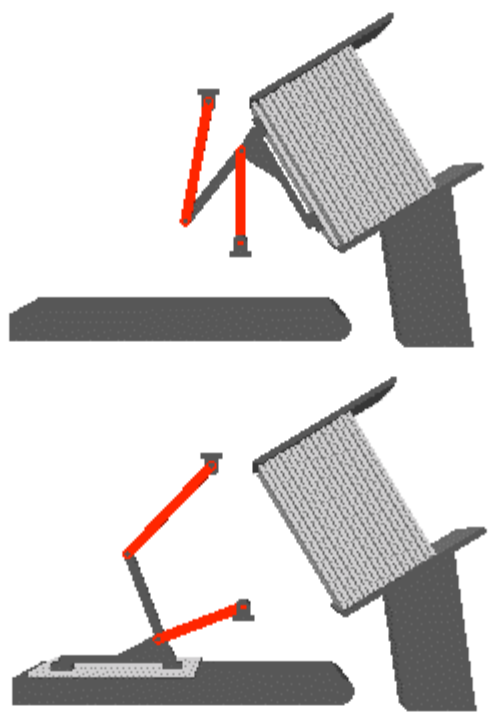
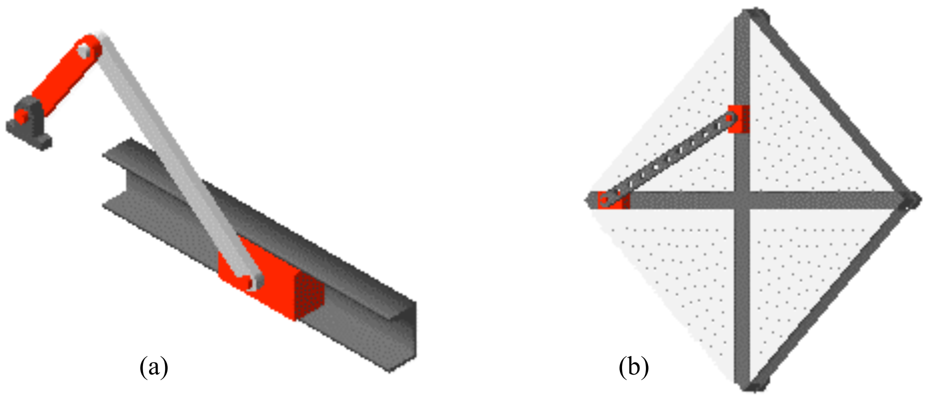


Figure 1.23 A four-bar pick-and-place mechanism

A robot could accomplish the same task but it would likely be three times the size and driven by two or three motors and programmable controllers. The robot inertia may limit speed of operation whereas the moving links in this design can be very light as the only motor needed to drive it is attached to the frame and the mechanism may be fully or partially balanced. Balancing refers to designing links with counter-masses to minimize their rotational inertia.

Another example of a four-bar mechanism is the automobile hood linkage discussed previously and shown in Figure 1.11b. In this case the input force is applied to the coupler link, which is integrally attached to the hood. The linkage is designed to guide the hood through precise positions and orientations in open and closed positions.

If we replace one of the joints in a four-bar linkage with a slider (another f_1 type joint), we generate the familiar piston-crank mechanism, Figure 1.24a, used in a conventional automobile engine. In an engine, the piston or the slider is the input link, the coupler link is the *connecting* rod and the output is the crank. When we employ such a mechanism for pump applications, the crank typically serves as the input and the reciprocating pumping motion of the piston is the output. If we convert the pin connection between the crank and the frame to a sliding connection, we have the *double-slider mechanism* shown in Figure 1.24b. One application of this mechanism is to change the direction of the input motion imparted to one of the sliders by 90 degrees.



(a) (b)

Figure 1.24 (a) Four-bar mechanism with one slider; the familiar piston-crank mechanism and (b) four-bar with two sliders and two pin-joints.

1.6.2 FIVE-BAR MECHANISMS

A chain of five binary links connected by f_1 type joints, Figure 1.25a, has two degrees of freedom. One can intuitively verify this by holding link 2 in a fixed position and noting that link 5 can still rotate about its pin connection with the ground. If you then fix link 3, while holding link 2 fixed, link 5 becomes immobile. You have essentially frozen two single degree of freedom joints, joint a between the frame and link 2 and joint b connecting links 2 and 3 before all the other links (links 4 and 5) are rendered immobile. Therefore the linkage has two degrees of freedom. You can easily verify this result by Gruebler's equation with $n = 5$, $f_1 = 5$ and $f_2 = 0$. Two actuators are required to control joint c (called the knee). You can attach the two actuators to any two joints but practical experience strongly suggests that you should connect actuators to grounded joints whenever possible, in this case joints a and e, to minimize the moving mass and inertia. If you want to reduce the number of actuators, then you could mechanically couple the two input links, 2 and 5, by a pair of gears. Figure 1.25b shows a geared five-bar mechanism. Note that links 2 and 5 are now ternary links. Link 2 has three joints; it connects to links 1 and 3 with pin joints and to link 5 with a gear connection. Recalling that a gear connection is a two degree of freedom joint, we compute the degrees of freedom of the mechanism as follows.

$$F = 3(5 - 1) - 2(5) - 1 = 1$$

A geared five-bar mechanism generates intricate motions without adding significant complexity.

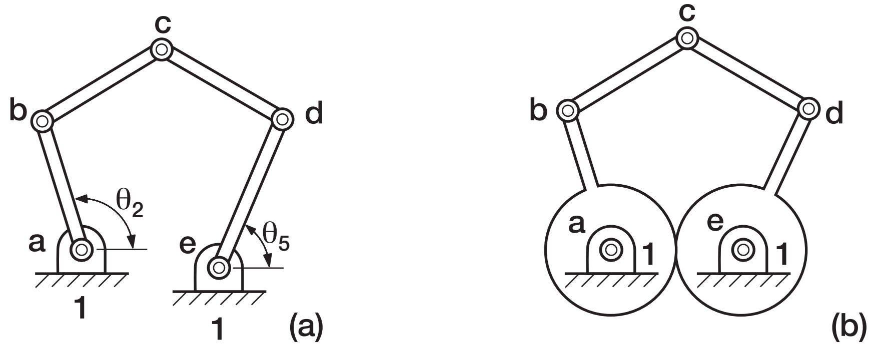


Figure 1.25 (a) A five-bar mechanism with two degrees of freedom.
(b) A geared five-bar mechanism with one degree of freedom.

Six-Bar Mechanisms
There are only two six-link mechanisms composed of binary and ternary links that have a single degree of freedom.

1.6.3 SIX-BAR MECHANISMS

Six-bar mechanisms are used in many applications particularly when higher pair connections are deemed undesirable. The prosthetic knee mechanism shown in Figure 1.2 is an example of a single degree of freedom six-bar mechanism. The motorcycle suspension mechanism shown in Figure 1.1 is a different type of six-bar which is equivalent to a four-bar mechanism with a shock-absorber added. There are numerous ways in which one can combine six links (binary, ternary, quaternary) links with f_1 type joints to generate planar six-bar mechanisms. If we restrict ourselves to single degree of freedom six-bar mechanisms, then the number of possible combinations is limited. If we further restrict ourselves to only binary and ternary links, then the number of such combinations reduce to only two (Figure 1.26):

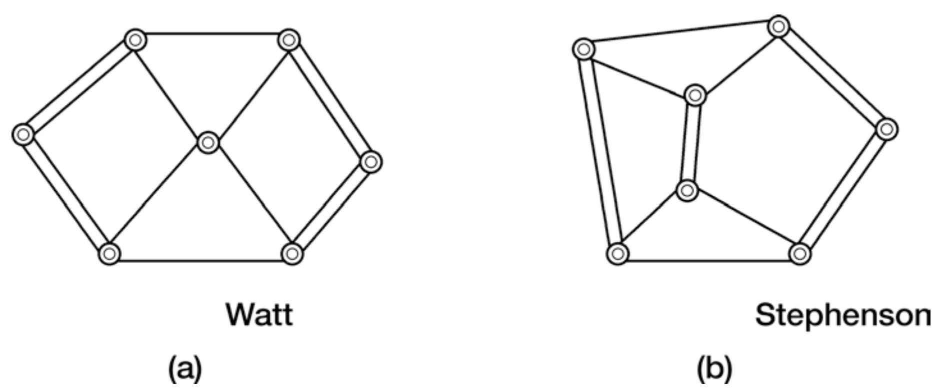


Figure 1.26 Two distinct one-degree of freedom six-link kinematic chains

a) Watt six-bar: Two ternary links and four binary links where the two ternary links are directly connected to each other.

b) Stephenson six-bar: Two ternary links and four binary links where the two ternary links are separated by a binary link.

There are five distinct ways in which one can assign the frame link to these two kinematic chains (Figure 1.27). In the case of the Watt six-bar kinematic chain there are two distinctly different connections:

a) Watt-1 six-bar: Ground link (fixed link) is a binary link.
b) Watt-2 six-bar: Ground link (fixed link) is a ternary link.

Likewise, in the Stephenson chain, there are three distinctly different link combinations.

a) Stephenson-1: Fixed link is connected to two ternary links.

b) Stephenson-2: Fixed link is connected to one ternary link and one binary link.
c) Stephenson-3: Fixed link is a ternary link.

Although these classifications may seem to be purely academic at this stage they can reveal the uniqueness or patentability of a specific kinematic chain in the context of other existing designs. In all five types of mechanisms shown in Figure 1.27, $n = 6, f_1 = 7$ and $F = 3(6 - 1) - 2(7) - 0 = 1$.

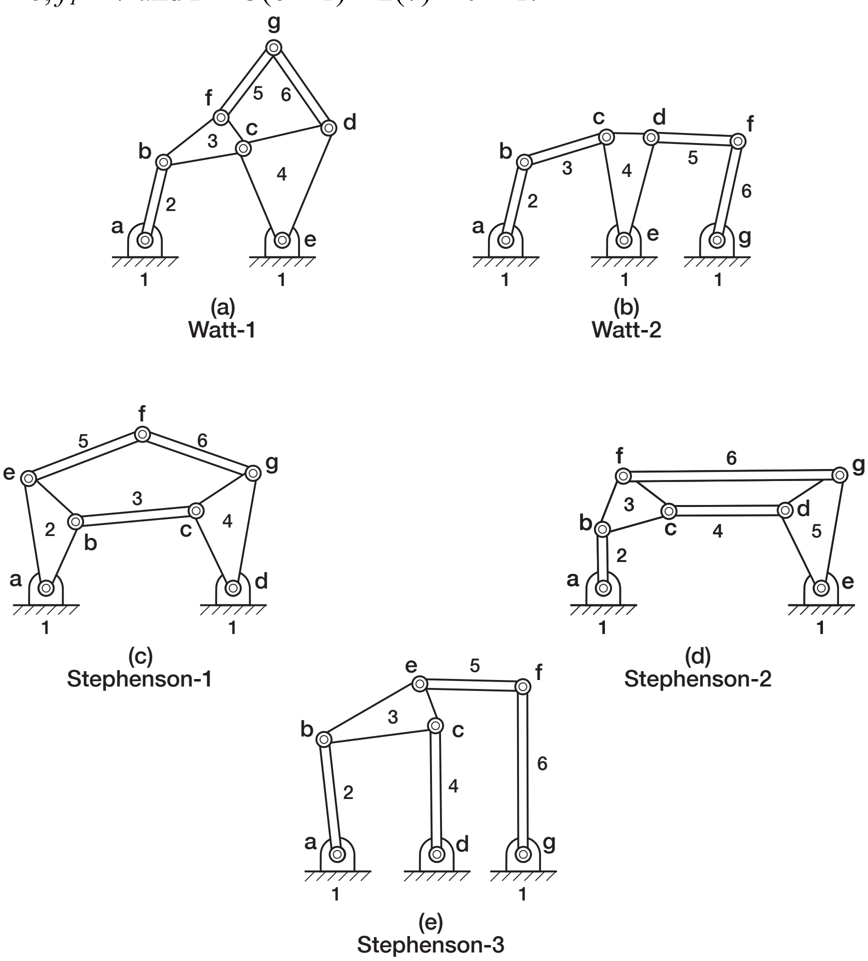


Figure 1.27 Five distinct six-bar mechanisms.

1.7 EXAMPLES:

1.7.1 TRENCH-HOE

The trench hoe shown in Figure 1.28 has 12 links (consider the cab as the ground link), 12 pin joints, and three slider joints (the piston-cylinder combinations). If you counted only 11 pin connections, look more carefully at point Q in the figure. Three links are connected by the same pin constraint. There are two pin joints at Q, one connecting links 9 and 10, the other connecting links 10 and 11. In general, if m links are joined by a single revolute joint, the number of pin joints at that common connection is (m – 1)

The number of degrees of freedom of the trench hoe is therefore

$$F = 3(12 - 1) - 2(15) = +3$$

Thus the trench hoe linkage requires three input motions to determine the position of all the links relative to the cab. These are supplied by the three hydraulic cylinders that are attached along the boom.

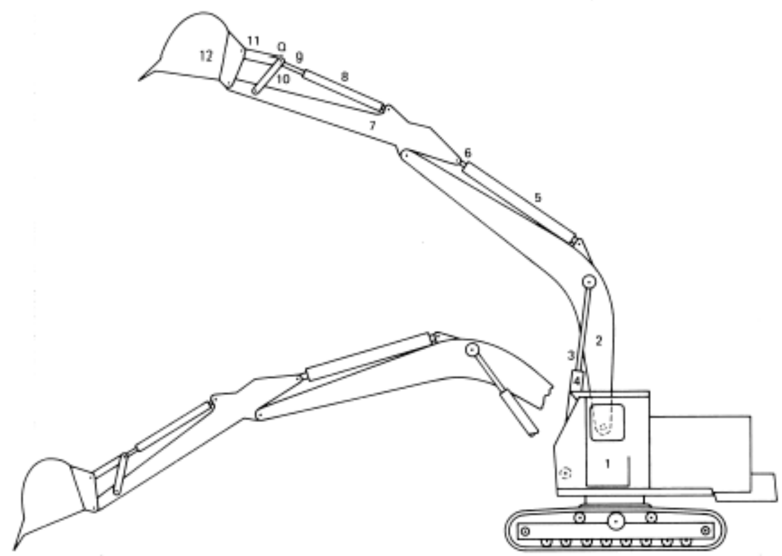


Figure 1.28 Trench-hoe mechanism

1.7.2 CASEMENT WINDOW MECHANISM

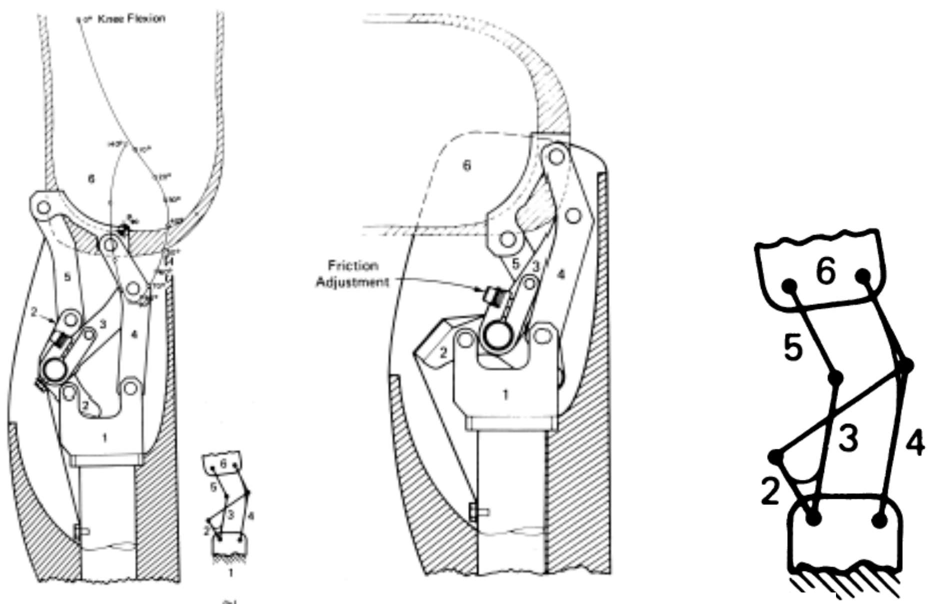
Returning to Figure 1.13a shows one of the popular casement window operator mechanisms in the 90-degree and 30-degree positions, respectively. A casement window must open 90 degrees outward from the sill and be at sufficient distance from one side to satisfy the egress codes and on the other side it should provide access to the outside of the window pane for cleaning. Also, the force required to drive the linkage must be reasonable for hand operation. If you have used one of these at home, you know that it requires a single hand-crank and therefore has $F = 1$. The first step in analyzing any mechanism is to draw its kinematic or skeleton diagram. Figure 1.13b shows the kinematic diagram (sketch) for the casement window linkage.

Notice that there are six links, five pin joints, one slider joint, and one pin-in-slot joint in this sketch. Note also that one loop of the mechanism contains a slider-crank linkage (1, 5, 4, 6). Connected to the slider crank is a bar and a roller

(2, 3), which provides the input for opening and closing the window. The kinematic diagram simplifies the mechanism for visual inspection and, if drawn to scale, provides the means for further analysis. Remembering that the pin-in-slot is an *f1* joint with the no-slip assumption this mechanism has F = 3(5) – 2(7) – 0 = 1 degrees of freedom. What type of six-bar is represented in Figure 1.13? First, we search out the ternary links: Links 1 and 4 each connect to three other links and are not directly connected to one another. Referring to Figure 1.27, we determine that the casement window mechanism is a Stephenson-III six-bar.

1.7.3 PROSTHETIC KNEE MECHANISM

In the design of an external prosthesis for a through-knee amputee, it is desirable to duplicate the movement of the relative center of rotation between the thigh and the shank to maintain stability in walking. Figure 1.29 shows a six-bar mechanism designed for this purpose in two different positions. The zero-degree flexion (fully extended) position is shown in Figure 1.29a together with the trajectory of the *instant center* (see section 2.2 in the next chapter) of rotation of link 6, the artificial leg, with respect to link 1. The 90 degree flexion (bent knee) position is shown in Figure 1.29b. It’s easier to understand and analyze the underlying kinematics of this device by studying Figure 1.23c. Note that links 2 and 4 are the two ternary links in the six-bar chain with only two connections to the ground. This identifies it is a Stephenson-I six-bar mechanism. Basic knowledge of kinematic chains and degrees of freedom keeps you from misinterpreting link 4 as a binary link and counting a seventh link, incorrectly yielding two degrees of freedom.



(a) (b) (c)

Figure 1.29 Prosthetic Knee Mechanism. (a) Fully extended position, (b) Bent knee position and (c) Kinematic diagram

1.8 ADJUSTABLE MECHANISMS

In some applications, there will be a need to adjust the mechanism so that it functions a bit differently in response to a change in load, range of motion or some other parameters. For instance, the lawn sprinkler mechanism shown in Figure 1.10, is a simple *four-bar* mechanism where the sweep angle of the

Adjustable Mechanisms require an additional degree of freedom beyond what is required for the normal operation of the mechanism.

oscillating output link can be changed to fit the geometry of the lawn surface. The mechanism is thereby modified by relocating a ground pivot. Essentially, we change the length of one of the links (the distance between a set of pivots) to alter the motion characteristics of the mechanism. Of course, you must design-in necessary adjustments when you are creating such mechanisms so that the motion characteristics are altered as desired. The adjustment function by itself requires a degree of freedom that is arrested during normal operation. Otherwise it would have two degrees of freedom continuously. The adjustment can be physically achieved in a number of ways. A common approach is to add a sliding contact point that can be relocated and fixed with a screw mechanism. The screw is turned manually or with a small motor. The joint to be relocated is mounted on the nut which translates in response to rotation of the screw.

1.8.1 EXAMPLE: VARIABLE DISPLACEMENT ENGINE MECHANISM

The mechanism shown in Figure 1.30 varies the stroke of a piston engine to adjust power output. How the stroke linkage works is shown in Figure 1.31.

For each position, the lower end of a control link is adjusted along an arc prescribed by the control yoke shown. The top of the control link is connected to the main link which, in turn, connects to a component that plays the role of a conventional connecting rod. In essence, the result is an engine with variable crank-throw.

When control-yoke divergence from vertical is slight (Figure 1.31a) the main link is restricted in its movement, and the resulting piston stroke is small. As the control nut moves inward on its screw, the angle between the control yoke and "the axis of the cylinder" is increased. This causes the main link to move in a broader arc, creating a longer stroke. The angle between the control yoke and the cylinder axis varies between 0 and 70 degrees; the resulting stroke length varies from 1 in. to 4.25 in. The linkage is designed so that the compression ratio stays approximately the same, regardless of piston stroke.

The equivalent un-scaled kinematic diagram of this adjustable mechanism is also shown in Figure 1.30. Notice that there are nine links, nine pins, and two sliders in this sketch. Thus, $F = 3\,(8) - 2\,(11) = 2$.

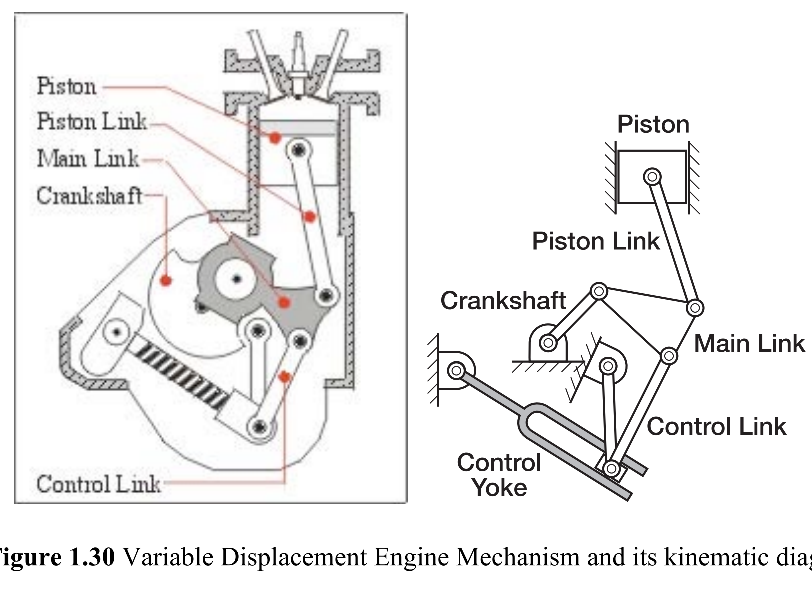


Figure 1.30 Variable Displacement Engine Mechanism and its kinematic diagram

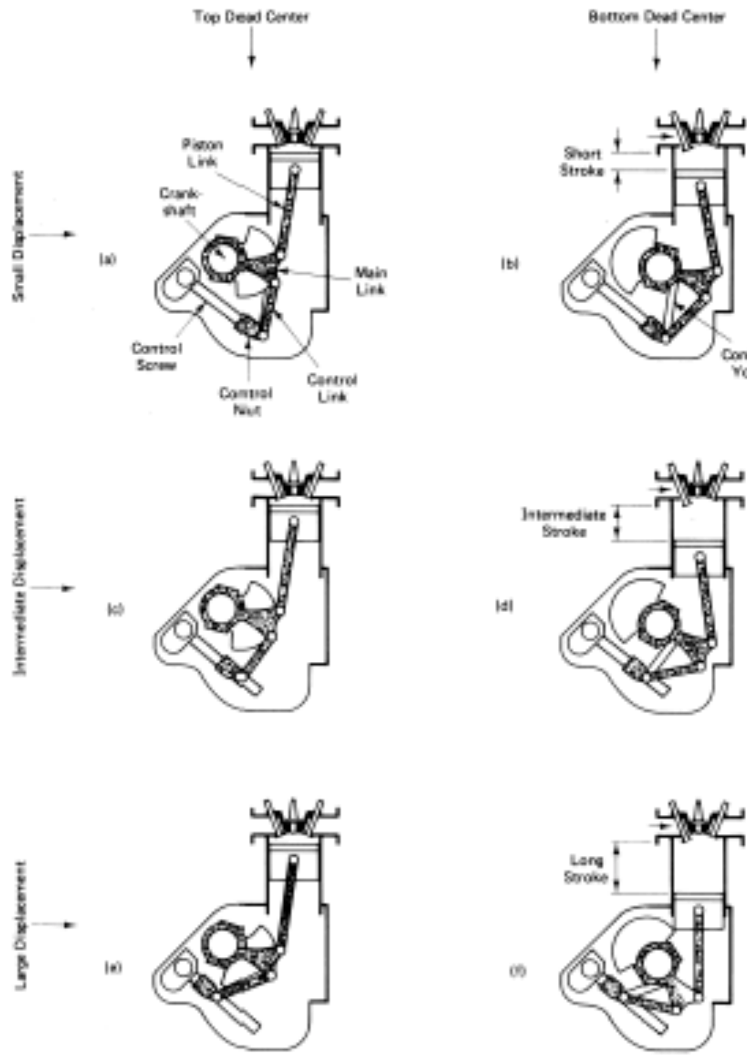


Figure 1.31 Variable displacement piston mechanism shown in various positions.

Another example of a mechanism that allows for adjustment is a phasing gearbox. Commonly used on rotary equipment, this gearbox uses a two-degree of freedom gear train, such as a planetary gear set. Instead of fixing the outer (ring) gear, that gear can be rotated by a worm gear drive. This allows adjustments in phasing or timing of two different parts of a system. Bevel gears can also be used, as shown in Figure 1.32 below.

Figure 1.32 Phasing gear reducer

1.9 UNDERSTANDING KINEMATIC FUNCTION OF MECHANISMS

We propose you consider the process described in Figure 1.33 when trying to understand how a complex mechanism functions. It is often difficult to figure out how a device operates even when a number of positions of the mechanism are shown. Sometimes, it is also difficult to distinguish between moving links and fixed brackets and the type of joints employed. Actually, you are already prepared to accomplish this analysis with the tools and techniques presented thus far.

Apply this progression to the following exercise. Figure 1.34 shows a Truck Bed Tilting Mechanism taken from the US patent 4,066, 296.

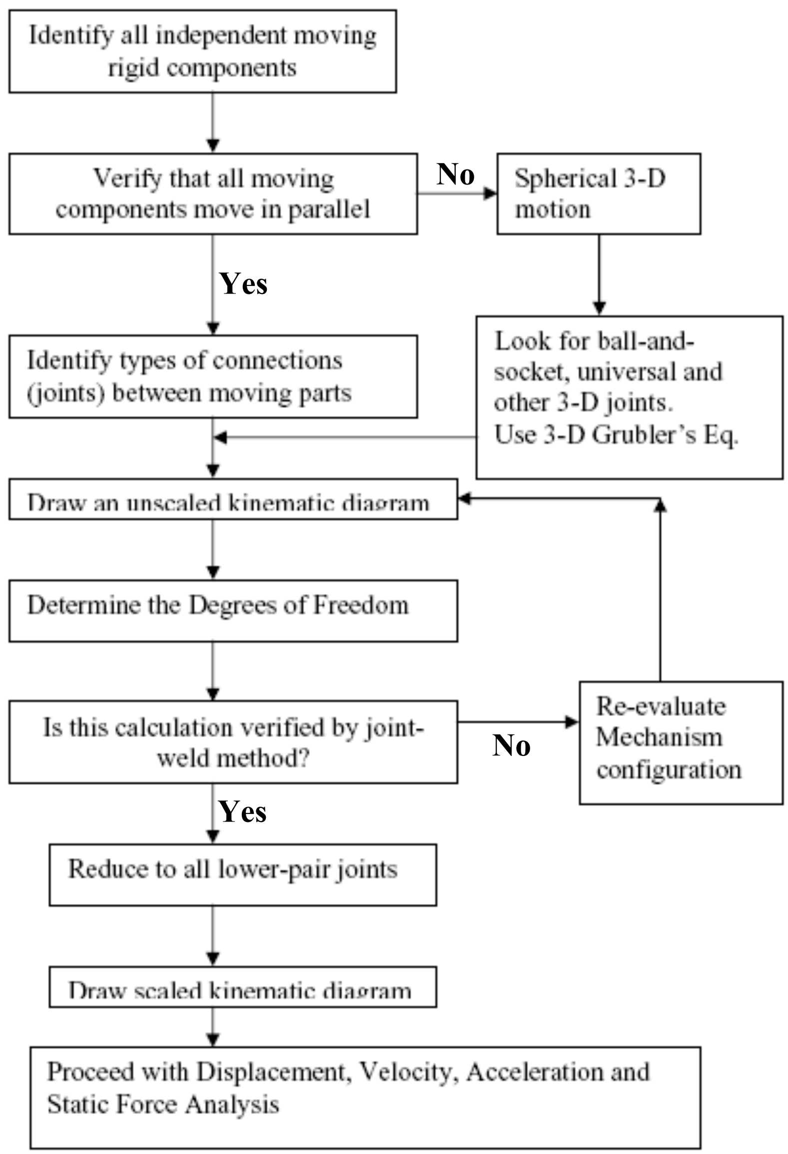


Figure 1.33 Identification of kinematic function of a complex mechanism.

1.9.1 TILTING TRUCK BED MECHANISM EXAMPLE

U.S patent #4,066,296 entitled "Truck Bed Tilting Mechanism" was awarded to Louis M. Ray Jr. in 1978. By examining Figure 1.34a and the kinematic diagram of the mechanism in Figure 1.34b, we see that the movement of the truck bed (36) is controlled by a linkage mechanism (18). A hydraulic cylinder (180) and the mechanism are shown in two positions (solid and dashed lines) in Figure 1.34a. As you can see, it is difficult to interpret these figures. We will use what we know of some basic tools: joint types, kinematic chains and degrees of freedom, to interpret the mechanism. For example, we know that for single degree of freedom chains, we must have at least four links if all the joints are lower pairs. Using Figure 1.33, let us study the bed tilting mechanism.

Step 1. Moving links appear to be the truck bed (36), the two sections of hydraulic actuator (180), triangular link (152), triangular link (150), link (154) and link (156). Including the frame of the truck as the ground link, this totals eight links.

Step 2. All links appear to move in parallel planes.

Step 3: All joints seem to be pin joints except the slider joint between the piston and the cylinder in the actuator (180).

Step 4: Figure 1.34b is an un-scaled diagram of the mechanism in this patent. Two sets of numbers are present, the numbering from the patent drawings and a simpler sequence starting with ground identified as link one. If we convert the sliding joint between the piston and the cylinder into a pin joint representation (they are both *f1* type joints), then a basic kinematic chain is generated as shown in Figure 1.34d.

Step 5: Verification of the analysis thus far is critical. Mistakes can be expected for complex systems. A degree of freedom evaluation helps identify any errors in identifying links and joints. Here $n = 8$, $f_1 = 10$ and $f_2 = 0$, thus $F = 3(7) - 2(10) = +1$.

Step 6: One degree of freedom is what we expected with only one hydraulic actuator. Note that the actuator is not connected to the truck frame with a pin joint.

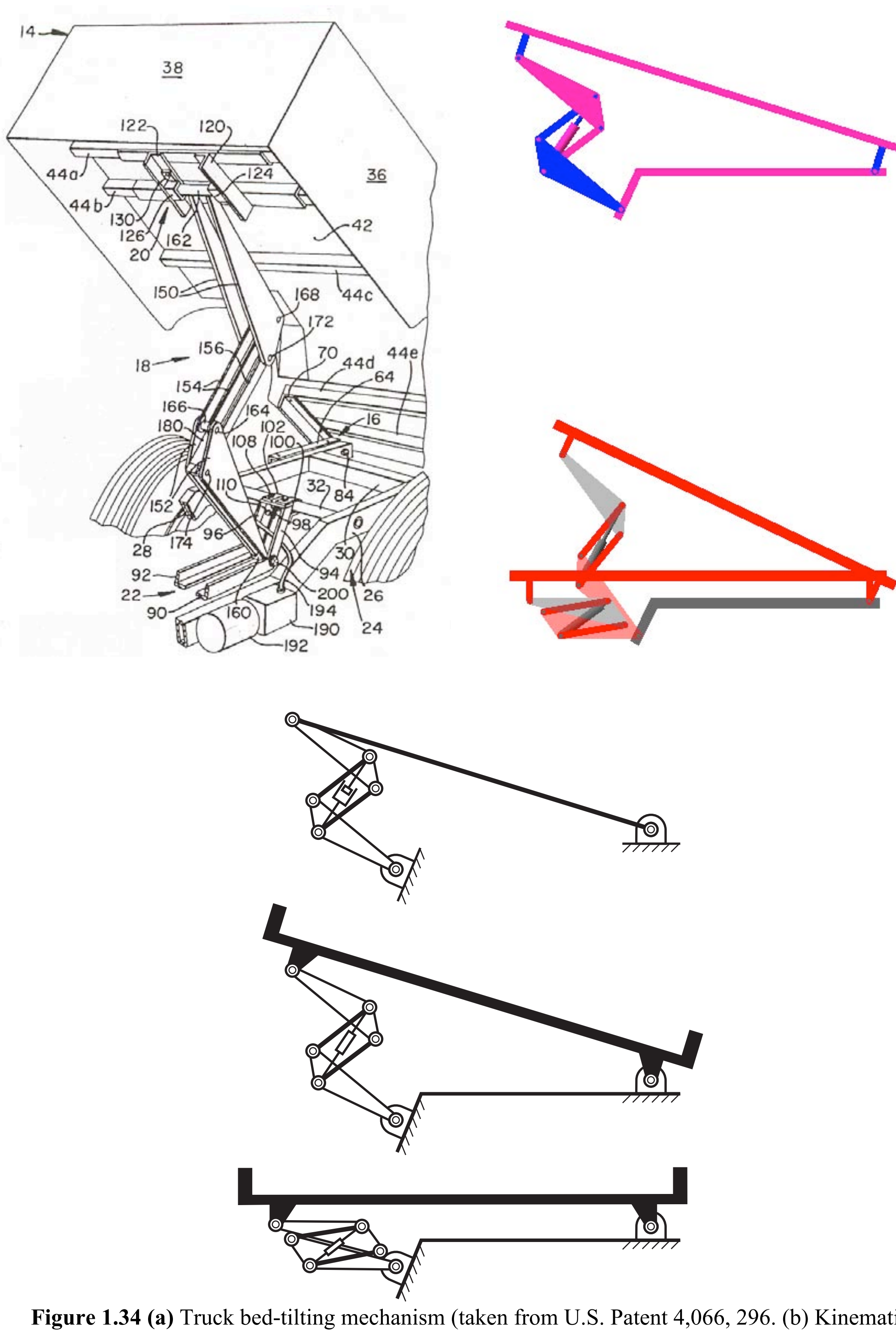


Figure 1.34 (a) Truck bed-tilting mechanism (taken from U.S. Patent 4,066, 296. (b) Kinematic diagram of the mechanism. (c) mechanism shown in the rest and full-tilt positions. (d) Basic kinematic chain of the mechanism.

1.10 SYNTHESIS VERSUS ANALYSIS

In any mechanical system that transmits movement and force from an input to an output we can identify at least three distinct entities: (i) the input (I), (ii) the mechanism (M) and (iii) the output (O). The input can be an electric motor, a hydraulic or pneumatic cylinder, or a manual input. The output can be a reciprocating motion with a clear forward to return stroke ratio, or some other desired motion. In an **analysis task**, the configuration and the components of the mechanism are known. We may perform kinematic or dynamic analysis of the mechanism to understand the exact behavior of the mechanism. In a synthesis task, the desired input motion (and forces) and the output motion (and forces) are known or prescribed. The goal of a **synthesis task** is to create a mechanism, that is, determine its exact configuration, and size various components such that it transmits the motion and forces from the input to the output precisely in a prescribed manner. In a **motion-programming task**, the mechanism characteristics and the desired output motion are known. The goal is to establish the exact nature of the input characteristics accounting for inertia effects and so on. Figure 1.35 depicts the essential differences among the three types of tasks.

Analysis: the process of determining the behavior of a given mechanism.
Synthesis: the process of creating a mechanism to yield a particular behavior.

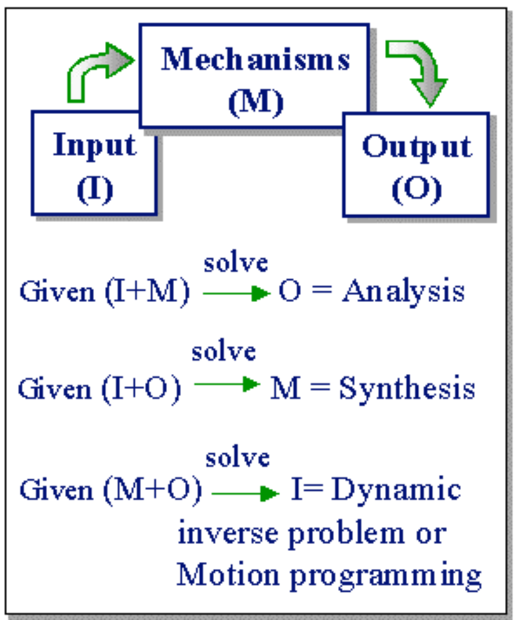


Figure 1.35 Synthesis, Analysis and Motion Programming tasks

1.11 VARIOUS STEPS IN MECHANISM DESIGN

Figure 1.36 shows an overview of the mechanism design process whether you are re-designing a mechanism to improve its functionality and/or performance, or creating a new mechanism. Various steps involved in designing a successful mechanism are briefly described below. Note that a mechanism may be a part of a bigger system such as the orbital-motion mechanism, Figure 1.5, in an orbital saw, or it can be a stand-alone system as in the pick-and-place automation mechanism shown in Figure 1.23. The steps described below apply to any mechanism.

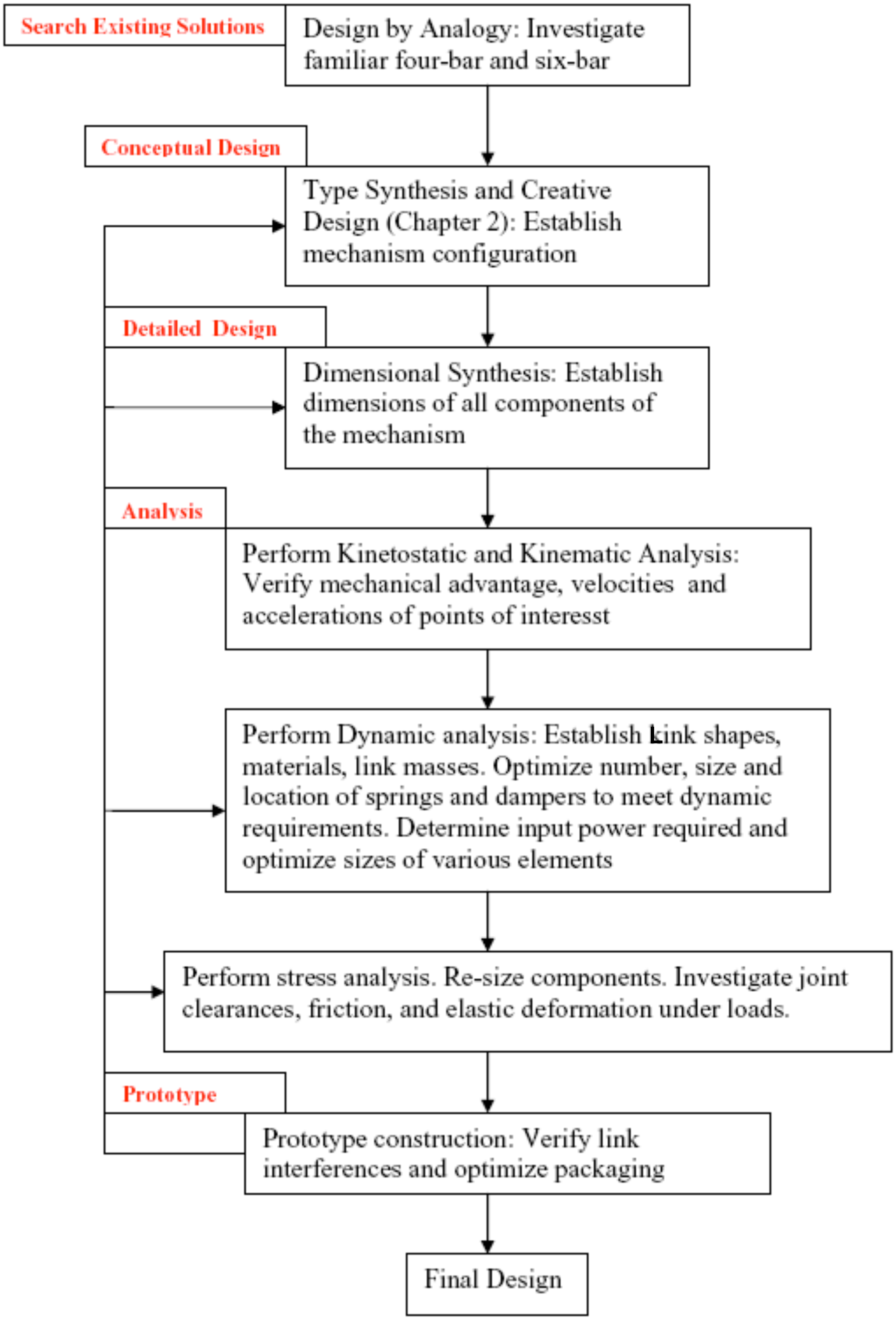
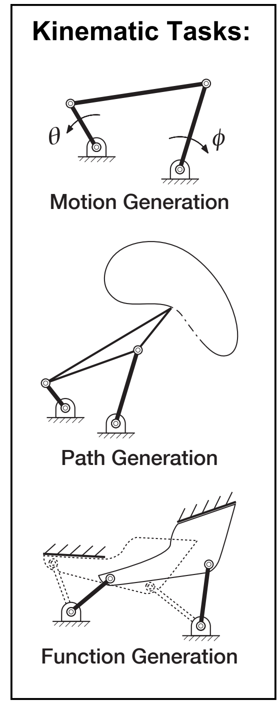


Figure 1.36 An overview of various steps in mechanism design process.

1.11.1 PROBLEM FORMULATION

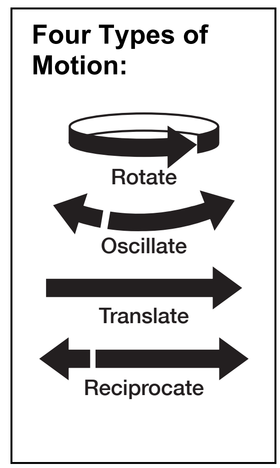
The first step in the design of a mechanism is to define the essential function of the device to be created. To do this you must formulate the design task in terms of fundamental motion requirements. These requirements can be posed in terms of basic kinematic requirements as one of the following:



(a) Guide a rigid body through successive positions and orientations in a two- or three-dimensional design space. Figure 1.11b shows a four-bar automobile hood linkage. The linkage must be designed to control the relative orientation between the hood and the car frame. The hood must not interfere with the frame of the car as it opens and must fit flush into the cavity in the car in the closed position. The location of the front of the hood and the angle of the hood relative to the frame of the automobile are critical. Such a task where the position and orientation of a rigid body are prescribed in the design requirements is called **rigid-body guidance** or **motion generation**.

(b) Trace a desired path of a point on one of the links of the mechanism. For instance the *four-bar* mechanism shown in the forklift truck of Figure 1.8c must be designed such that the point P of link 3 traces a vertical straight line of a certain height as link 2 is rotated through a certain angle. Such a task where the path traced by a certain point on a moving link is prescribed in the design requirements is called **path generation**.

(c) Coordinate the motion of an output link with respect to input. That is, the output must rotate, oscillate, translate or reciprocate through a certain angle or distance in response to either continuous rotation or translation of the input link. For instance, in designing a linkage for the lawn-sprinkler, we would specify that the output link oscillates back and forth 70 degrees in response to continuous rotation of the driver link. Such a task where the motions of two links must be coordinated is called **function generation**.



1.11.2 GENERAL DESIGN REQUIREMENTS

Most, if not all, mechanism design problems can be categorized into these three kinematic tasks: (a) *motion generation*, (b) *path generation* and (c) *function generation*. These three constitute the highest-level design goals, and the most basic functionality level tasks. In some cases, the kinematic function may be a combination of more than one of the three tasks. Say for instance there is a need to guide a rigid body along a prescribed path and then tilt it through a certain angle while another rigid body translates through a specified linear distance. In any case, several other design constraints and performance requirements constitute the entire design specification. These include:

1. **Compactness**: The choice of mechanism type may significantly influence the compactness of the mechanism, say in a stored position. Sometimes a mechanism with more links may be more compact than another with fewer links. The convertible-top mechanism (Figure 1.3) is a good example of compactness in the stored position.

2. **Ease of Maintenance**: This is largely a matter of the choice of joint type. Lower-pair joints (pin joints in particular) tend to hold the lubricant better and last longer.

3. **Low Power Consumption**: Reducing the inertia of the moving masses is an important factor. Proper application of dynamic elements (springs and flywheels) to store and release energy during different parts of the motion cycle can significantly influence the total and peak power requirements.

4. **Reconfigurabilty**: There are applications where you would want the mechanism to either self-adjust or adjust through external intervention to meet the changing demands of motion or load. The designer has to incorporate additional degrees of freedom (ex. Adjusting screws) in the mechanism which will be activated on demand.

5. **Robustness**: This term is sometimes misused to represent ruggedness. Its true meaning is being insensitive to variations. You would want the mechanism to function properly in spite of unavoidable variations; for instance dimensional variations arising from manufacturing, or assembly.

6. **Smooth Operation**: The mechanism should run without binding and must be easy to operate. In both cases, this depends again on the choice of joint types and other important factors such as transmission angles and mechanical advantage which influence how effectively the motion is transmitted from the input to the output. We will discuss these in greater detail in Chapter 2.

7. **Reliability**: As with any design, the number of moving parts directly impacts reliability of the system and the goal is always to keep that number to a minimum. Therefore, in the early phases of design it is important to choose the simplest mechanism that is appropriate. It is also very important to note that reliability is greatly influenced by the type of joints in the mechanism not simply the number of moving parts. Higher pair joints reduce the part count but at the expense of increased maintenance and decreased reliability.

8. **Cost**: The types of joints in the mechanism significantly influence the material and assembly cost and the cost of maintenance.

Other design requirements include speed reduction, orientation of the input and the output axis of motion, generation and coordination of the motion of multiple outputs in response to a single input, multiple inputs controlling the motion of a single output, etc.

1.11.3 SEARCH FOR EXISTING SOLUTIONS

Before embarking on creating a whole new mechanism to carry out the prescribed task, it is important to search for existing solutions since there is no point in "re-inventing the wheel". One useful step is to perform a broad patent search online at the United States Patent and Trademark office http://www.uspto.gov. It not only helps avoid duplicating what has been already done, but it also provides insight into how others have tried to address similar problems. The search can also motivate you to pursue a superior new design.

1.11.4 CONCEPT GENERATION

The first step in generating your mechanism concept is to abstract the problem at hand to a upper level and search for concepts developed for analogous problems. A good source to search for mechanism concepts is the book "Mechanisms, Mechanical Devices and Controls" edited by Nicholas Chironis. The book is also available on line at http://www.knovel.com

A systematic way to generate alternate mechanism concepts is called Type Synthesis. Type Synthesis refers to determining the *type* of mechanism to employ to perform the given task, which is, establishing the number and type of links, joints, and the manner in which they are connected. The number of kinematic degrees of freedom needed to generate a desired function must also be clear in the Problem Formulation. Type Synthesis techniques can be applied to enumerate all possible combinations of mechanisms that satisfy certain kinematic degrees of freedom and other design constraints. Some of these techniques allow you to start with an existing mechanism (a competitor's design) and the generate kinematically distinct topologies that perform the same function.

Concept Generation: Don't rush through the early stages of the design process. Making wise decisions at this stage is critical to achieve a good design.

1.11.5 CONCEPT EVALUATION

At this early stage of design, it can be more challenging to objectively evaluate alternate designs than it is to generate them. Since the dimensions, geometry, power requirements, etc. are not yet established, it is difficult to rank alternate concepts in terms of typical user requirements such as weight, compactness, low-power, energy efficiency, desired mechanical advantage, ease of maintenance, etc. However, some general guidelines can help at this stage. For instance, the type of kinematic joints employed can have a profound impact on maintenance cost and complexity. Sliding joints are not well suited in applications which expose the joints to dirt and other type of debris. Higher-pair connections, such as line contacts, tend to squeeze out lubricant, and therefore require extensive care in ensuring proper lubricant is available to keep the device running smoothly. For example valve trains in automobile engines have stringent demands on materials and heat treatment methods for the cams and still require a lubrication process in which the parts are submerged in a bath of oil. The type of links (binary, ternary), that are directly connected, can intuitively give some indication of the forces imparted to frame or sliding members. The decisions made in this early phase of design will have a profound impact on the product cost, quality, and time to market.

1.11.6 DETAILED KINEMATIC DESIGN

This step involves two distinct stages.

(i) Dimensional Synthesis
(ii) Kinematic Analysis

Once the type of mechanism is established, the next step is to establish the exact dimensions of various link elements, geometry of cam and follower surfaces, and the number of gear teeth if called for. This is called Dimensional Synthesis. Chapter 4 discusses analytical methods and computational tools that are available for designing linkages. Starting with the precise positions and orientation of the rigid body to be guided, the Dimensional Synthesis method generates numerous linkage solutions that can perform the kinematic task. Selecting what works best in a given application depends on many factors including:

(a) how the mechanism fits within the available design space without interfering with other parts of the system

(b) Transmission characteristics. How well the input forces are transmitted to the output? This can be evaluated by determining the minimum transmission angle, or pressure angle between the imparted output forces and the output displacements.

(c) Mechanical or Geometric Advantage. It addition to performing the kinematic motion, the mechanism may be required to amplify the input force (mechanical advantage) or amplify the input displacement (geometric advantage).

(d) The peak velocities, and peak accelerations of various members of the mechanism must be analyzed before establishing the final design.

A synthesis software program for designing linkages, called LINCAGES is available through the companion website. The companion CD has another software program for synthesizing cams and followers, called CAMSYN.

1.11.7 DETAILED DYNAMIC DESIGN

Establishing the type and dimensions of a mechanism to satisfy the kinematic requirements is a very critical early step. The dynamic behavior of the mechanism must be understood and optimized in order to develop a good practical solution. The mass and inertia of various elements of the mechanism must be taken into consideration in evaluating their dynamic performance. Dynamic analysis must be carried out to determine:

(a) **Forces** in various joints so that proper bearings or bushings can be employed.

(b) **Torque-Speed** (or Force-Displacement) characteristics of the input(s) so that proper motors are selected to drive the mechanism.

(c) **Location, Size, and Characteristics** of dynamic elements such as springs and dampers to optimize the velocity and acceleration characteristics of the mechanism.

(d) **Clearance Effects** on various joints such as impact and backlash.

(e) **Elastic Deformation** of links, particularly in high-speed mechanisms that are subjected to large inertial forces.

(f) **Friction** in various joints and its effect on overall performance.

(g) **Forces and Moments** imparted to the frame, called shaking forces and shaking moments. The mechanism must be balanced to minimize these forces and thereby minimize the vibrations imparted to the supporting structure.

Analytical prototypes can be readily created in commercial software programs such as ADAMS and it can be accessed through the companion website. This and other similar highly sophisticated commercial software programs allow the user to create a three-dimensional model of the mechanism and perform realistic simulations of its dynamic behavior. Available software programs can easily determine all the force calculations listed above. Basic concepts in dynamic analysis are covered in Chapter 5.

1.11.8 PROTOTYPE CONSTRUCTION

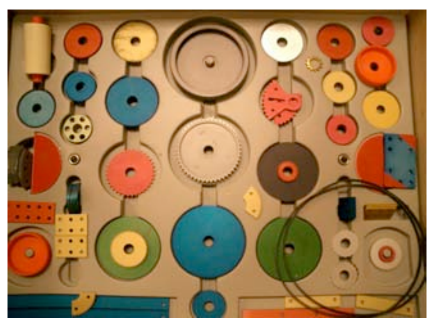
Once a mechanism is designed dimensionally, it is highly recommended that you create a proof-of-concept prototype. Constructing a physical prototype provides greater insight into proper assembly of the mechanism to avoid link interference, connecting the driving link to the actuator etc. There are many simple methods for creating a physical prototype:

(a) Constructing a linkage system using individual links that can be easily machined to desired shapes using a laser-cutting machine or other conventional machines. Typically, laser-cutting machines can cut most plastics (Acrylic, ABS, etc) of up to 2 inch thickness and also create holes for pin joints. Most machines can read at least one of the typical CAD file formats. A few typical formats are IGES, DXF, DWG, SAT, STP, and STL. In these cases, the part drawings (CAD) can readily be transformed into physical links. Shoulder-bolts from a local hardware store can be used to create hinge joints. Figure 1.37 shows a student mechanism design project prototype created from laser-cut links and shoulder-bolts.

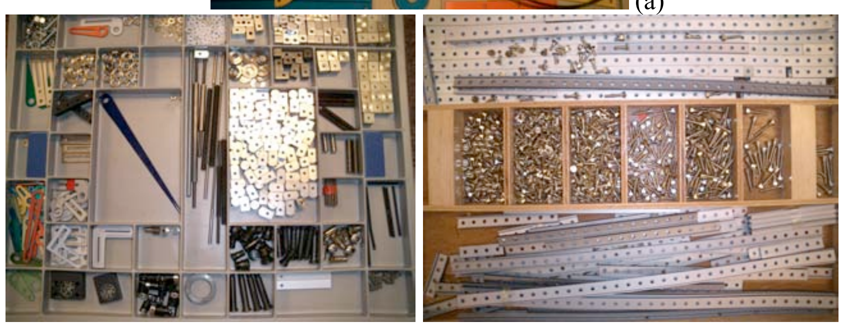
Figure 1.37 Prototype constructed from laser-cut acrylic links in a student mechanism design project.

(b) Rapid Prototyping from a CAD model is another method of transforming part files of links or cams directly into physical parts. Again, either off-the-shelf shoulder-bolts can be used for pin connections, or a plastic version of the same can be created using the same rapid prototype machine.

(c) Many commercial vendors offer an assortment of gears, racks, motors, etc. for prototyping. Several companies, such as W. M. Berg and Stock Drive Products, offer an assortment of metallic components such as gears (spur, bevel, etc), pulleys, timing belts, cranks, etc. (Figure 1.38a) as well as modular beams for constructing a frame to mount the mechanism (Figure 1.38b). These components can be used to create functional prototypes. One such prototype of a bevel differential is shown in Figure 1.39a. A prototype of a student mechanism constructed from these construction kits is shown in Figure 1.39b. Fischer Technics offers a variety of plastic construction kits including straight and curved links, joints, tracks, sliders, motors, reduction gears, etc. Figure 1.40 shows one such assortment.



(a)



(b)

Figure 1.38 Standard components from a commercial vendor (Stock-drive products) find use in quick prototype construction (a) various rotational components and (b) various structural components.

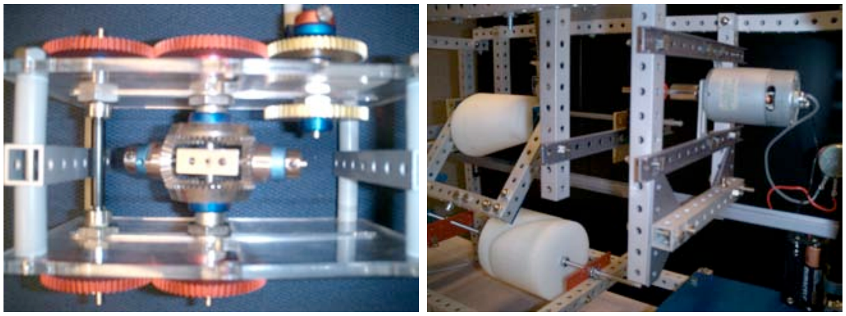


Figure 1.39 Commercial kinematic teaching and project aids. (a) Working model of an automotive differential. (b) A student-project prototype constructed from off-the-shelf components shown in Figure 1.38.

Figure 1.40 Off-the-shelf plastic components from a commercial vendor (Fischer Technics) consisting of links and joints for quick prototype construction.